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The Informativeness of Sentiment Types in Risk Factor Disclosures: Evidence from Firms with Cybersecurity Breaches

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ABSTRACT: This study examines the degree to which Loughran and McDonald (LM) word lists are informative at the item level of SEC filings, such as risk factors (RF) and management's discussion and analysis (MDA) disclosures in 10-X reports. In this context, we explore if sentiment types are informative when associated with other material events, namely cybersecurity breaches. Our results support the assertion that sentiment types, beyond positive and negative, are informative at the individual disclosure item level, as tested in the RF and MDA sections. We also find that investors respond to different types of sentiment between RF and MDA. We find an economically significant estimated average economic impact of $469 million/firm. We further contribute to the literature by applying novel statistical methods that advance empirical accounting literature.

Data Availability: Data are available from the public sources cited in the text.

Keywords: cybersecurity; market reactions; cumulative abnormal returns; trading volume; sentiment analysis; textual analysis; risk factors; management's discussion and analysis; PLS-SEM; genetic matching.

I. INTRODUCTION

Cybersecurity risks are a significant factor when investing (Securities and Exchange Commission (SEC) 2011), and a large body of research confirms the significant economic impact of cybersecurity breaches on a firm's value (Goel and Shawky 2009) and the market (Spanos and Angelis 2016). As such, the SEC has recently issued guidance on disclosing cybersecurity risks and incidents (Securities and Exchange Commission (SEC) 2018, 2011). Unfortunately, the current corporate disclosure literature tends to focus on positive and negative sentiment types only (Dougal, Engelberg, Garcia, and Parsons 2012; Mayew and Venkatachalam 2012; Huang, Zang, and Zheng 2014; Berkman, Jona, Lee, and Soderstrom 2018) or overall disclosure tones (Huang, Teoh, and Zhang 2013), bypassing other available sentiment types (Loughran and McDonald 2011; Bodnaruk, Loughran, and McDonald 2015), such as

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uncertain, litigious, constraining, strong modal, medium modal, and weak modal language, which could provide additional informative clues in the disclosure for investors.

The paucity of studies that use sentiment types beyond positive and negative motivates this study, which seeks to answer a few empirical questions. First, are Loughran and McDonald (LM) word lists informative at the item level of a 10-K report, such as risk factors (RF) and management's discussion and analysis (MDA) disclosures for understanding the related market reactions? Second, besides material internal control weaknesses, are sentiment types informative when associated with other material events, such as cybersecurity breaches? Third, are different sentiment types informative to investors besides positive and negative sentiments?

This study aims to answer these empirical questions with an event study methodology to investigate the market effects of RF and MDA disclosures of 378 breached firms through sentiment analysis. Our results support the assertion that sentiment types, beyond positive and negative, are informative at the individual disclosure item level, as tested in the risk factors and management's discussion and analysis sections. We also find that investors respond to different types of sentiment between RF and MDA. This finding is beneficial for management when preparing disclosure content.

Furthermore, after controlling for internal control weaknesses and restatements, our empirical evidence shows associations between cumulative abnormal returns and various sentiment types in RF and MDA disclosures. The results not only answer our second research question affirmatively that sentiment types are informative for cybersecurity breaches, but they also positively support our third research question that sentiment types other than positive and negative are informative. The robustness test further reveals that sentiment types in RF are significant after 10 days of the report date and 60 days after the report date in MDA, with improved model predictability. The statistically significant associations show economic significance in RF ($469 million) and MDA ($503 million). Using partial least squares structured equation modeling (PLS-SEM), the alternative testing displays meaningful relationships between sentiment types and trading volume (*TVOL*) and the bid-ask spread (*SPREAD*). The results suggest the informativeness and value relevance of the sentiment types in corporate disclosures. The results of the PLSpredict (Shmueli et al. 2019) procedure indicate the out-of-sample predictive models for *TVOL* and *SPREAD*.

We contribute to the disclosure and sentiment analysis literature in the following ways. First, our study extends the empirical application of LM word lists (Loughran and McDonald 2011) to the report item level within the complete 10-K reports, specifically RF and MDA. Second, this study's results contribute to the growing debate on the informativeness of risk factor disclosures on cybersecurity. Third, they also provide empirical evidence that other sentiment types are informative beyond positive and negative. The informativeness of sentiment types may motivate future research to explore the application and usefulness of sentiment types in corporate disclosure and other content. Fourth, this study provides insight into the predictive power and the longevity of sentiment types in disclosures between RF and MDA on market reactions. Fifth, this research builds on prior accounting information systems (AIS) and accounting research to address corporate disclosure of a critical AIS issue, cybersecurity. Specifically, we contribute methodologically to the accounting and AIS literature by introducing PLS-SEM, a lesser known but rigorous statistical method, as our alternative test. Additionally, we adopt the genetic matching algorithm, a superior matching technique (Diamond and Sekhon 2013), to validate our results. In summary, not only is this study relevant to AIS by broadening the literature relating to corporate disclosures, cybersecurity, and sentiment analysis, but the empirical results and the advanced statistical techniques used in this study will also benefit future accounting research in this area and beyond.

The following section provides an overview of the literature and hypothesis development. Section III describes the sample selection. Section IV explains the data analysis and results. Finally, we discuss the conclusions, implications, and limitations in Section V.

II. LITERATURE REVIEW AND HYPOTHESES

Risk Factor Disclosures

A large body of literature on risk factor disclosure has existed since the SEC's introduction of firm-specific disclosure requirements. Releases Nos. 33-8591 and 34-52056 described external firm-specific risks, such as market and regulatory risks, and internal risks, including organizational, performance and capital risks (Securities and Exchange Commission (SEC) 2005). Mandated registrants disclose these risk factors in annual and quarterly reports (SEC 2005). Driven by the mission to protect investors and the implicit goal of reducing information asymmetry, the SEC advocates for investors by requiring public companies to disclose meaningful financial and other relevant and material information to the public.[1] The SEC's mandate to disclose risk factors in registrants' periodic filings to enhance their report contents

[1] https://www.sec.gov/Article/whatwedo.html

to better inform investors and markets (SEC 2005) has stimulated many disclosure research studies. The empirical results from recent studies have alleviated concerns that risk disclosures may be boilerplate. For example, Kravet and Muslu (2013) reveal that changes in textual risk disclosures are positively related to stock return volatility, trading volume, and the dispersion of analyst forecasts and revision volatility. Similarly, Campbell, Chen, Dhaliwal, Lu, and Steele (2014) conclude that risk disclosure is informative. The five subcategories of risks they created demonstrated statistical significance in market reactions, indicating that the market perceived the specific risk information conveyed through risk disclosures (Campbell et al. 2014).

In his seminal study, Li (2006) assesses the risk sentiment of corporate annual reports by counting the frequency of the word *risk* and its different permutations. The results in Li (2006) show that higher risk sentiment, which increases uncertainty and investor risk perception, is linked to lower future stock returns and earnings. Conversely, Wang, Rees, and Kannan (2013) provide evidence that firms with action-oriented terms (i.e., strong modal) in their security risk disclosures suffer significantly fewer adverse market reactions than firms without action-oriented terms in their disclosures. Furthermore, Li, He, and Xiao (2018) showed that the frequency of risk disclosure is positively associated with investment efficiency. The results are even more evident when the tone of the MDA sections is more positive in a large sample of 10-K and 10-Q reports (Feldman, Govindaraj, Livnat, and Segal 2010). However, most of these prior studies have generalized the sentiment of textual content to positive and negative. The full range of sentiment types created by Loughran and McDonald (2011) and Bodnaruk et al. (2015), which provides a broader and richer understanding of textual content, has not been fully utilized or researched.

Cybersecurity Risk Disclosures Research

Consistent research interest in disclosure is essential and justifiable, as corporate disclosure is critical for an efficient capital market (Healy and Palepu 2001). As technology advances, reporting requirements evolve (Filzen 2015). In 2011, advancements in internet technology prompted the SEC to address the emerging threat of cybersecurity risks. The guidance explicitly requires registrants to include cybersecurity risks and incidents in Item 1A (i.e., the risk factors) of the 10-K. The SEC also requires discussing or disclosing cybersecurity risks or events in MDA, description of business, legal proceedings, financial statement disclosures, and controls and procedures (SEC 2011). Our study focuses on the sentiment analysis of RF and MDA disclosure.

Unlike risk factor disclosure (SEC 2005), which has attracted the attention of researchers, newer disclosure guidance from the SEC on cybersecurity risks and incidents (SEC 2011, 2018) did not generate a similar level of interest from researchers. Although the risk of cybersecurity incidents is one of the most significant factors when making an investment decision (SEC 2011), the literature specific to the market effects of cybersecurity risk disclosure is limited. One study by Amir, Levi, and Livne (2018) identified a different relationship between risk reporting among firms that self-reported their cybersecurity breach and firms forced to disclose after outside parties reported cybersecurity incidents. Another cybersecurity risk disclosure study explored investment attractiveness for firms with prior cybersecurity attacks (Frank, Grenier, and Pyzoha 2019). A different study examined the ability to predict future cybersecurity incidents based on disclosure content (Wang et al. 2013). In research on market reactions to information data breaches in the United States from 2005 to 2017, Michel, Oded, and Shaked (2020) found an adverse market reaction in short-window trading just before the breach announcement and a marginally significant positive market reaction in a longer window (i.e., 20 days) following the announcement. Similarly, Richardson, Smith, and Watson (2019) found no significant relationship between breached firms and cumulative returns beyond short windows (−1,3).

Informativeness of Cybersecurity Risk Disclosures

Using textual analysis of information in risk disclosures, Wang et al. (2013) revealed that firms with action-oriented terms in their security risk disclosure experience a significantly smaller negative market reaction than firms without action-oriented terms in their disclosures. The results suggest that the market perceives the sentiment disclosed by firms in their SEC filings and behaves accordingly. More recently, Li, No, and Wang (2018) had similar results in their untabulated findings regarding market reactions after cybersecurity incidents, indicating that the market reacts differently based on the content of cybersecurity risk disclosure (i.e., informativeness). Furthermore, Amir et al. (2018) showed that the negative impact on the equity value of firms withholding information about cybersecurity incidents is more significant than that of firms disclosing incidents voluntarily. Michel et al. (2020) also revealed a negative mean abnormal return before a breach announcement and a positive mean abnormal return after the announcement, indicating the possibility of news leakage. Finally, Berkman et al. (2018) compiled a cybersecurity awareness score based on specific keywords and used only positive and negative word lists (Loughran and McDonald 2011) as control variables to investigate the impact on firms' market valuation. They found a positive relationship between market valuation and awareness score.

On the contrary, Hilary, Segal, and Zhang (2016) found a significant relationship between firms' negative abnormal returns and the count of cybersecurity risk-related expressions in RF and MDA after a cybersecurity incident. However, the results lack economic significance, and they concluded that cybersecurity risk disclosure is boilerplate. Their textual analysis consisted of only counting the frequency of the phrase *cybersecurity risk* and related permutations based on practitioner guidance and popular press (Hilary et al. 2016). Our study postulates that additional sentiment types from the disclosure content may provide more insight into the relationship between cybersecurity disclosure and market reactions.

Word Lists and Sentiment Types

Based on a computational linguistic program to classify investment messages (i.e., buy, hold, or sell) on internet stock message boards, Antweiler and Frank (2004) created a measure called bullishness to predict market volatility and stock returns. Similarly, Das and Chen (2007) developed an algorithm to extract sentiment from stock message boards and find a relationship between sentiment and stock values, stock index levels, volumes, and volatility. However, the algorithms deployed in these studies vary and thus lack consistency and comparability.

In addition to machine learning, researchers use word lists to analyze sentiment. One of the most important steps in measuring sentiment or tone is deciding on the specific dictionary to measure the attributes (Loughran and McDonald 2016). Earlier researchers in accounting and finance typically used Harvard's General Inquirer (GI) and Henry (2008) because of their availability. However, Loughran and McDonald (2011, 2016) attribute the use of these dictionaries to the fact that earlier studies fail to find meaningful incremental information content in corporate disclosures (Kothari, Li, and Short 2009; Tetlock 2007) when using GI, which is unsuitable in the business context. As such, Loughran and McDonald (2011) examined word usage in 10-Ks from 1994 to 2008 through the creation of word lists (collectively called LM word lists), which have become the *de facto* dictionary in accounting and finance research (Kearney and Liu 2014).

The inclusion of modal words in LM word lists is necessary. English modal verbs have multiple semantic meanings, also known as polysemous (Taylor 2003). Linguistic scholars categorize modal verbs as epistemic and deontic. Epistemic verbs, such as *might*, reflect the writer's attitude, commitment, or belief. In contrast, deontic verbs, such as *shall* and *can*, describe the necessity, ability, or possibility of the actions (Gelderen 2003). However, modal verbs like *may, should, will, would, could*, and *must* operate as epistemic and deontic (Gelderen 2003). Hence, Loughran and McDonald (2011) extend Jordan's (2003) modal verbs and categorize them as strong, medium, and weak modals to indicate the level of confidence in the 10-K reports. Combining the financial constraint word list (Bodnaruk et al. 2015), the eight sentiment types available in the LM word lists are positive, negative, uncertain, litigious, constraining, strong modal, medium modal, and weak modal.

These sentiment types are linguistic characteristics that provide a semantic clue of the disclosures expressed by firms' management in SEC reports. Positive words, such as *achieve, efficient*, or *improve*, are unilateral in a tone expressing optimism and confidence. Negative words are specific to business terminology, conveying adversity or difficulties in the business. Examples of negative words are *restate, misstatements*, or *penalties*. Words for uncertainty denoting ambiguity and hesitation include *depend*, *contingency*, or *indefinite*. Litigious words, such as *claimant*, *deposition*, or *regulation*, reflect a tendency for legal contests or a more regulated environment. Modal words categorized as strong, medium, or weak modal words indicate the levels of confidence in the subject of the sentence. Examples of modality words and their related categories are *always* (strong), *must* (strong), *likely* (medium), *should* (medium), *seldom* (weak), and *depend* (weak) (Loughran and McDonald 2011). Finally, words for constraining type, such as *required*, *obligation*, or *imposed*, imply a firm's financial limiting or constricting conditions (Bodnaruk et al. 2015). Appendix B provides some examples of sentiment types manifested in RF and MDA disclosures.

Despite the richness and availability of the eight sentiment types, the extant textual analysis literature on corporate disclosure primarily concentrates on positive and negative sentiment types or overall disclosure tones. The positive and negative sentiment from the LM word lists are used to find a direct relationship between pessimistic journalistic tone from newspaper columns and negative market returns the next day (Dougal et al. 2012). Studies using only the positive and negative words of the LM word lists show a positive association between managers' emotional state on earnings conference calls and concurrent stock returns (Mayew and Venkatachalam 2012), a significant link between abnormal positive tone in earnings press releases and negative future earnings and cash flows in a long-window timeframe (Huang et al. 2014; Huang et al. 2013), and a strong predictive association between opinions from peer investors on social media investment platforms and future stock returns and earnings surprises (Chen, De, Hu, and Hwang 2014). More recently, Affuso and Lahtinen (2019) found that investors' sentiment (i.e., positive and negative) from Twitter posts has economic significance for daily stock returns.

Testing beyond positive and negative sentiment, Gao, Calderon, and Tang (2020) selected the litigious sentiment type as a dependent variable in a study to investigate the linguistic characteristics of companies' cybersecurity disclosure and company performance. Their analysis found that smaller firms include more litigious words in their cybersecurity disclosure. However, they did not find a significant relationship between cybersecurity incidents and the volume of litigious words. However, their study did not involve testing litigious sentiment and market reactions.

Sentiment Types and Risk Proxies in A Specific Report Section

Since Item 1A (i.e., risk factors) intends to disclose a firm's information about risks (i.e., uncertain sentiment type), disclosure sentiment is generally unfavorable or negative (Campbell et al. 2014). In addition to testing sentiment as positive or negative, this study is the first to utilize all eight sentiment types (Bodnaruk et al. 2015; Loughran and McDonald 2011) to investigate the disclosure content in RF and MDA of selected SEC reports. Including a broader range of sentiment types in the model provides a more meaningful understanding of the relationships between risk disclosures and market reactions beyond the two polar sentiment types. Although no empirical precedents exist in applying all sentiment types to analyze cybersecurity disclosures, there is no reason to question the relevance of LM word lists (Loughran and McDonald 2011) in this application. LM word lists are the *de facto* business dictionary (Kearney and Liu 2014), developed from the compilation of SEC annual reports after the introduction of RF by the SEC (2005).

To establish sentiment types as semantic clues of risks disclosed by management in SEC reports, we follow Campbell et al. (2014) to assess the association of sentiment types to the following five risk proxies: litigation risks (*SIZE*), financial risks (*LEV*), other systematic risks (*BETA*), profitability (*DNI*), and idiosyncratic risk (*VWRET*). Given that this study's sample consists of firms that have experienced cybersecurity breaches, we should expect an association between the sentiment types and the related risk proxies before disclosure. Therefore, we predict that sentiment types are meaningful and informative in risk disclosure from firms that have experienced cybersecurity incidents.

H1: Sentiment types are informative as risk proxies in risk factor disclosure.

Sentiment Types and Market Reactions

Just as important as including a range of sentiment types in analyzing risk disclosure, understanding investors' reactions in perceiving these sentiment types in risk disclosure is paramount. Management can better prepare disclosure content by understanding the relationships between sentiment type and related market reactions. Similarly, retail investors can learn from the sentiment of the disclosure and better prepare for the market volatility that a cybersecurity incident may bring.

The extant risk disclosure literature has shown a positive association between short-window market reactions and 10-K filings, suggesting that risk disclosures are informative to investors (Hope, Hu, and Lu 2016). More specifically, Richardson et al. (2019) examine the cumulative abnormal returns in short (−1,3) and long windows (−1,21) between breached firms and matched nonbreached firms[2] around the disclosure date. Based on a binary variable to operationalize breached firms, their results show that the differences in the cumulative daily returns, cumulative excess returns, and cumulative abnormal returns between breached and nonbreached firms are statistically significant. Without a textual or sentiment analysis, their study leaves room for this paper to offer possible explanations. Prior research shows that perceived risks are associated with market reactions (Campbell et al. 2014; Kravet and Muslu 2013). Following our first hypothesis that sentiment types are strongly associated with risk proxies, we predict that different sentiment types are significantly associated with market reactions, specifically cumulative abnormal returns around the SEC report dates.

H2: The sentiment types of risk factor disclosures are significantly associated with cumulative abnormal returns surrounding the SEC report release date.

Related research shows that the market reacts differently according to action-oriented terms (Wang et al. 2013) or based on the content of cybersecurity risk disclosure (Li, No, and Wang 2018), indicating a connection between market reactions and the specific risk information content disclosed. However, the paucity of research utilizing all sentiment types of the LM word lists (Loughran and McDonald 2011) limits our understanding of the relationships between various sentiment types beyond the positive and negative market reactions. Therefore, extending work that shows a significant relationship between litigious sentiment and company performance (Gao et al. 2020), we predict that the market reacts differently to various sentiment types between unique sections of the SEC reports, namely, RF and MDA.

[2] Nonbreached firms are based on both Propensity Matched and Market Value-Matched methods (Richardson et al. 2019).

H3: Cumulative abnormal returns are significantly associated with different sentiment types between disclosure sections (i.e., RF and MDA).

III. SAMPLE

Cybersecurity Breach Data

This study analyzes the relationships between the sentiment types of risk factors (i.e., Item 1A) and the associated market reactions based on a sample of publicly traded firms with cybersecurity incidents. We collect cybersecurity incidents from the Privacy Rights Clearinghouse[3] (PRC), a nonprofit organization established in 1992 to educate and advocate individuals' privacy rights. PRC aggregates breach data from state attorney generals, government agencies, Data Loss DB, Databreaches.net, the National Association for Information Destruction, the Open Security Foundation, the Personal Health Information Privacy, and the media. In addition, PRC provides a specific date and magnitude for each breach, such as the number of records lost or stolen, an incident description, and the organization involved. PRC has been used in finance literature (Michel et al. 2020), accounting literature (Hilary et al. 2016), information systems literature (Li, No, and Wang 2018), and legal literature (Romanosky, Hoffman, and Acquisti 2014).

Because cybersecurity events may occur randomly, the confounding event effect requires attention due to the strong market reaction to earnings release announcements (Li and Ramesh 2009). Therefore, to isolate the impact of cybersecurity disclosure on market reactions outside of known confounding events, we remove 59 observations from 31 companies for which their cybersecurity disclosures coincided with their earnings announcements outside of their SEC filing dates. See Table 1 for the final sample selection.

SEC Reporting

Consistent with prior studies, we extract registrants' filings from the SEC's Electronic Data Gathering, Analysis, and Retrieval (EDGAR) database by matching the company names in the PRC with central index keys.[4] Considering the specific SEC cybersecurity disclosure guidance issued in 2011, we include breach firms before and after the published guidance to assess differences in the disclosures. Due to the sporadic filing of the 8-K[5] reports and the limited content contained within, we focus our textual analysis on risk factor disclosures and management's discussion and analysis reported on 10-Ks and 10-Qs. We select the SEC filings for our sentiment analysis based on the date of breached firms' first filing at least 30 days after the PRC's reported incident date. Before including an SEC report in our textual analysis, we deploy a Python program that uses *regular expression*[6] (RE) to check for keywords[7] that imply or signify cybersecurity content within the disclosure. The RE search failed to find cybersecurity domain keywords in SEC reports related to 20 incidents. Six other incidents had missing corresponding SEC filings and were thus removed from the final sample. The final number of observations for the current study is 378 from 199 unique U.S. companies from 2009 to 2018 (see Table 1), which consists of 308 10-Ks and 70 10-Qs[8] for sentiment analysis.

Market Reaction Measures

Market reactions are the dependent variable in the current study. Given the event-specific nature of this study, we analyzed the cumulative abnormal return using the event study software by Wharton Research Data Services (WRDS)[9] based on a final sample size of 378 observations. Our parameters include the Fama and French (1996) three-factor model as the market risk model, with 100 days in the estimation window requiring a minimum of 70 observations and a two-day gap before the event window of one day before and one day after the SEC filing date (−1,1), which is the short window widely used in the literature (Filzen 2015; Filzen, McBrayer, and Shannon 2023; Hammersley, Myers, and

[3] http://www.privacyrights.org
[4] We extracted from Compustat a list of company names (CONM), CIK number (CIK), and ticker symbol (TIC) between 2008 to 2019. Using Fuzzy Lookup in Excel, we matched company names from PRC with company names from Compustat. The unmatched company names were researched and matched manually with the corresponding TIC or discarded as nonpublic companies.
[5] Only 17 8-K or 4 percent of the total initial observations met the selection criteria of timing and keywords.
[6] Regular expression is a highly specialized programming language embedded inside Python available through the *re* module. Please visit https://docs.python.org/3/howto/regex.html for more details on the usage and specifics about regular expression in Python.
[7] Adopted the cybersecurity risk keywords and phrases from Li, No, and Wang (2018).
[8] Many of the 10-Qs meet the timeline criteria for selection but lack the cyber keywords for inclusion in our analysis.
[9] Event Study by WRDS is a paid-subscription service provided by the University of Pennsylvania. For more details related to the specific cross-sectional significance test of the event study, please visit https://wrds-www.wharton.upenn.edu/

TABLE 1

Sample Selection of Cybersecurity Breaches and Breached Firms by Industry

Selection Criteria	No. of Breaches	No. of Breached Firms
Breaches from 2009 to 2018 in BSF, BSR, MED, and BSO	5,813	4,974
Less: Firms that are nonpublic companies	(4,945)	(4,439)
Less: Firms not found in CRSP or Compustat	(349)	(286)
Less: Observations with confounding events	(59)	(31)
Less: Foreign corporations	(23)	(10)
Less: Duplicate records	(14)	—
Total preliminary observations with CRSP and Compustat	423	208
Less: Incidents related to 8-K filings	(17)	(1)
Less: Missing or incomplete SEC filings	(26)	(8)
Total final observations with all data	378	199
Breached observations by industry		
BSF	89	42
BSO	110	72
BSR	84	51
MED	95	34
Observations by year		
2009	19	18
2010	42	31
2011	39	31
2012	34	31
2013	49	35
2014	63	50
2015	32	29
2016	39	34
2017	39	35
2018	22	17

Industry coding is provided by PRC.

Shakespeare 2008; Wang et al. 2013). Unless otherwise mentioned, market reaction assessment and control variables are obtained from CRSP and Compustat (Appendix A provides a comprehensive list of the variable definitions).

Disclosure Sentiment

We adopted LM word lists[10] with eight sentiment types from the Notre Dame Software Repository for Accounting and Finance (SRAF).[11] A Python program calculates the value for each sentiment type based on the number of words matched by sentiment type in the LM word lists divided by the total word count of the disclosure type of the filing (i.e., RF or MDA) and multiplied by 100 for analysis. We apply the sentiment analysis on RF and MDA of the selected SEC reports containing cybersecurity keywords described earlier.

IV. DATA ANALYSIS AND RESULTS

Descriptive Statistics

The final sample of 378 incidents was fairly distributed across the four industries categorized by PRC. Business other (BSO) had the highest concentration of incidents (29.1 percent or 110 incidents), followed by healthcare, medical

[10] For a complete list of the words of each sentiment category, please visit https://sraf.nd.edu/loughranmcdonald-master-dictionary/. The moderate modal was added after 2011.

[11] The Notre Dame Software Repository for Accounting and Finance is located at https://sraf.nd.edu/

TABLE 2

Descriptive Statistics of Selected Variables (n = 378)

Variables	Mean	Median	Std. Dev.	0.25	0.75
POS_{RF}	0.985	0.968	0.300	0.801	1.172
NEG_{RF}	4.076	4.010	0.834	3.598	4.467
UNC_{RF}	3.213	3.234	0.532	2.909	3.517
LIT_{RF}	1.650	1.619	0.545	1.286	1.941
WMD_{RF}	2.151	2.154	0.466	1.899	2.453
MMD_{RF}	0.428	0.415	0.137	0.338	0.514
SMD_{RF}	0.356	0.348	0.142	0.260	0.438
CNT_{RF}	1.098	1.085	0.298	0.882	1.273
FOG_{RF}	20.349	20.243	2.315	19.290	21.257
POS_{MDA}	0.831	0.791	0.266	0.626	0.998
NEG_{MDA}	1.462	1.367	0.582	1.060	1.846
UNC_{MDA}	1.382	1.360	0.382	1.141	1.577
LIT_{MDA}	0.718	0.625	0.403	0.421	0.884
WMD_{MDA}	0.329	0.321	0.122	0.238	0.390
MMD_{MDA}	0.282	0.273	0.101	0.212	0.329
SMD_{MDA}	0.210	0.199	0.088	0.143	0.260
CNT_{MDA}	0.690	0.691	0.228	0.528	0.836
FOG_{MDA}	17.990	18.457	2.558	17.126	19.622
CAR	0.002	0.001	0.053	−0.017	0.019
$TVOL$	1.481	0.912	1.774	0.544	1.622
$SPREAD$	0.067	0.029	0.153	0.018	0.061
$VWRET$	0.035	0.035	0.067	0.017	0.074
$STDSRV$	0.019	0.017	0.011	0.012	0.022
BTM	0.534	0.399	0.643	0.223	0.668
$SIZE$	9.727	9.747	1.837	8.430	11.195

Definitions of all variables are in Appendix A.

providers, and medical insurance services (MED) at 25.1 percent, business financial and insurance services (BSF) at 23.5 percent, and business retail/merchant-including online retail (BSR) at 22.2 percent. As shown in Table 1, cybersecurity incidents reached their highest level in 2014, with 63 incidents, and dropped below 40 incidents per year for the next few years.

Table 2 provides interesting insights into the sentiment disclosed in RF and MDA disclosures from breached firms. Since cybersecurity breaches are generally considered negative news (Amir et al. 2018; Frank et al. 2019; Li, No, and Wang et al. 2018), we expect negative sentiment (*NEG*) to be the most dominant among the eight sentiment types captured in the current study. *NEG*[12] scored an average of 4.076 percent and a median of 4.010 percent, followed by uncertain (*UNC*) at 3.213 percent, weak modal (*WMD*) at 2.151 percent, and litigious (*LIT*) at 1.650 percent, whereas positive (*POS*) and constraining (*CNT*) sentiment averaged 0.985 percent and 1.098 percent, respectively. Our results are directionally comparable to the seminal textual analysis study by Loughran and McDonald (2011), who also reported a higher percentage of mean negative sentiment (1.39 percent) than positive sentiment (0.75 percent). Given that our sample contains only disclosures from companies with cybersecurity incidents, a mean *NEG* of four times the positive sentiment is unsurprising.

The average firm's market capitalization from this sample is \$16.76 billion ($e^{9.727}$), with a median of \$17.11 billion ($e^{9.747}$), which is consistent with prior research (Jasa 2019; Richardson et al. 2019). Table 3 displays the pairwise Pearson correlation matrix of the selected exogenous and endogenous variables tested in this study using a two-tailed test of significance. Among the significant correlations, positive sentiment (*POS*) was negatively correlated with negative

[12] We follow Loughran and McDonald (2011) in using simple negation. We treat a positive word as a negative word when it is preceded, within three words, by one of the negation words (*no*, *not*, *none*, *neither*, *never*, and *nobody*).

TABLE 3

Pairwise Pearson Correlation Matrix by Disclosure Type

Panel A: Pairwise Pearson Correlation Matrix of the Risk Factors Variables in the Current Study

Variables	1	2	3	4	5	6	7
1. POS_{RF}	1						
2. NEG_{RF}	−0.186***	1					
3. UNC_{RF}	0.004	0.496***	1				
4. LIT_{RF}	−0.310***	0.144***	−0.159***	1			
5. WMD_{RF}	0.128**	0.528***	0.775***	−0.141***	1		
6. MMD_{RF}	−0.036	−0.054	0.061	−0.018	−0.162***	1	
7. SMD_{RF}	0.211***	−0.194***	−0.054	−0.211***	−0.037	0.229***	1
8. CNT_{RF}	−0.173***	−0.111**	0.142***	0.107**	0.165***	0.181***	0.268***
9. FOG_{RF}	0.035	0.032	−0.114**	0.200***	−0.245***	0.283***	−0.028
10. *CAR*	−0.009	−0.141***	−0.079	−0.001	−0.024	−0.056	−0.042
11. *TVOL*	−0.079	0.024	−0.010	−0.104**	0.050	−0.150***	0.005
12. *SPREAD*	−0.032	−0.048	0.004	−0.153***	0.018	0.134***	0.136***
13. *VWRET*	−0.010	−0.039	−0.065	0.036	−0.022	0.113**	−0.009
14. *STDSRV*	−0.081	−0.012	0.069	−0.243***	0.087*	−0.013	0.163***
15. *BTM*	−0.183***	−0.143***	0.052	0.005	−0.192***	0.178***	0.088*
16. *SIZE*	−0.100*	−0.036	−0.086*	0.236***	−0.237***	0.031	−0.211***

Variables	8	9	10	11	12	13	14	15
1. POS_{RF}								
2. NEG_{RF}								
3. UNC_{RF}								
4. LIT_{RF}								
5. WMD_{RF}								
6. MMD_{RF}								
7. SMD_{RF}								
8. CNT_{RF}	1							
9. FOG_{RF}	−0.108**	1						
10. *CAR*	0.060	−0.028	1					
11. *TVOL*	−0.035	−0.133***	0.006	1				
12. *SPREAD*	0.064	−0.097*	0.031	0.032	1			
13. *VWRET*	0.011	0.114**	0.091*	−0.128**	−0.133***	1		
14. *STDSRV*	0.062	−0.186***	−0.100*	0.380***	0.389***	−0.218***	1	
15. *BTM*	0.112**	−0.011	0.067	0.043	0.224***	−0.039	0.366***	1
16. *SIZE*	−0.212***	0.304***	−0.035	−0.226***	−0.419***	0.007	−0.400***	−0.017

*, **, *** Denote 10 percent, 5 percent, and 1 percent two-tailed statistical significance levels, respectively.
Panel A presents a bivariate correlation matrix of the main variables in this study. n = 378.

(continued on next page)

(*NEG*, −0.186), litigious (*LIT*) (−0.310), and constraining (*CNT*) (−0.173) but positively related to weak modal (*WMD*) (0.128) and strong modal (*SMD*) (0.211). In addition, a meaningful and statistically significant correlation exists between cumulative abnormal returns (*CAR*) and *NEG* (−0.141). On the other hand, trade volume volatility (*TVOL*) shows a significant negative correlation with *LIT* (−0.104), *MMD* (−0.150), and *FOG* (−0.133).

Sentiment Types and Risk Proxies

Our first hypothesis predicts that the sentiment types depicted in risk factor disclosures are associated with a firm's risk proxies. Cybersecurity risks are not only IT risks but are also becoming enterprise risks (Clinton 2017) that jeopardize a firm's financial and operational performance. The SEC recognized the significance of cybersecurity risks and their

TABLE 3 (continued)

Panel B: Pairwise Pearson Correlation Matrix of Management Discussion and Analysis Variables in the Current Study

Variables	1	2	3	4	5	6	7
1. POS_{MDA}	1						
2. NEG_{MDA}	0.064	1					
3. UNC_{MDA}	0.080	0.374***	1				
4. LIT_{MDA}	0.003	0.360***	0.215***	1			
5. WMD_{MDA}	−0.043	0.144***	0.449***	0.331***	1		
6. MMD_{MDA}	0.171***	−0.012	0.375***	0.120**	0.383***	1	
7. SMD_{MDA}	0.208***	−0.062	0.061	0.221***	0.263***	0.215***	1
8. CNT_{MDA}	−0.018	0.290***	0.262***	0.289***	0.407***	0.257***	0.207***
9. FOG_{MDA}	0.151***	0.226***	0.267***	0.216***	0.052	0.214***	0.133***
10. *CAR*	−0.061	0.028	−0.105**	−0.053	−0.064	−0.107**	−0.095*
11. *TVOL*	−0.246***	−0.072	−0.094*	−0.064	0.036	−0.109**	−0.011
12. *SPREAD*	−0.122**	0.109**	−0.029	−0.086*	−0.075	−0.085*	0.000
13. *VWRET*	0.026	0.005	−0.009	0.000	0.018	−0.030	0.048
14. *STDSRV*	−0.172***	0.123**	0.035	−0.100*	−0.025	−0.132**	−0.019
15. *BTM*	0.017	0.431***	0.276***	0.109**	−0.070	−0.050	−0.051
16. *SIZE*	0.226***	−0.035	0.128**	0.003	−0.027	0.138***	0.058

Variables	8	9	10	11	12	13	14	15
1. POS_{MDA}								
2. NEG_{MDA}								
3. UNC_{MDA}								
4. LIT_{MDA}								
5. WMD_{MDA}								
6. MMD_{MDA}								
7. SMD_{MDA}								
8. CNT_{MDA}	1							
9. FOG_{MDA}	0.232***	1						
10. *CAR*	−0.025	0.043	1					
11. *TVOL*	−0.041	−0.153***	0.006	1				
12. *SPREAD*	0.031	0.030	0.031	0.032	1			
13. *VWRET*	−0.041	−0.015	0.091*	−0.128**	−0.133***	1		
14. *STDSRV*	−0.040	−0.027	−0.100*	0.380***	0.389***	−0.218***	1	
15. *BTM*	−0.104**	0.088*	0.067	0.043	0.224***	−0.039	0.366***	1
16. *SIZE*	−0.166***	0.007	−0.035	−0.226***	−0.419***	0.007	−0.400***	−0.017

*, **, *** Denote 10 percent, 5 percent, and 1 percent two-tailed statistical significance levels, respectively.
Panel B presents a bivariate correlation matrix of the main variables in this study. n = 378.

adverse impact on performance in systems security, finance, operations, and legal and regulatory compliance. As such, we predict that sentiment types in risk factor disclosures are informative regarding firms' litigation risks (*SIZE*), financial risks (*LEV*), other systematic risks (*BETA*), profitability (*DNI*), and idiosyncratic risk (*VWRET* and *STDSRV*).

We adopt the design from Campbell et al. (2014) and operationalize the sentiment types (*SENT*) as $POS_{Type,it}$, $NEG_{Type,it}$, $UNC_{Type,it}$, $LIT_{Type,it}$, $WMD_{Type,it}$, $MMD_{Type,it}$, $SMD_{Type,it}$, $CNT_{Type,it}$, and $FOG_{Type,it}$, where *Type* represents risk factor or management discussion and analysis, *i* represents the firm, and *t* represents the fiscal year. As such, we estimate the following regression equation to test H1:

$$SENT_{Type,it} = \beta_0 + \beta_1 SIZE_{it} + \beta_2 BTM_{it} + \beta_3 VWRET_{it} + \beta_4 LEV_{it} + \beta_5 STDSRV_{it} + \beta_6 BETA_{it} + \beta_7 DNI_{it} + \beta_8 ANALYSTS_{it} + \beta_9 INSTOWN_{it} + \text{year effects}_t + \text{industry effect}_i + \varepsilon_{it} \quad (1)$$

Overall, all eight sentiment types exhibit a rich and statistically significant association with the magnitude of the risk proxies (Table 4). First, *SIZE*, a proxy for litigation risk, was positively associated with LIT_{RF} (0.121, p = 0.001) and FOG_{RF} (0.551, p = 0.035) but negatively related to POS_{RF} (−0.045, p = 0.012) and WMD_{RF} (−0.081, p = 0.003).

TABLE 4

Multivariate Analysis of Sentiment Type Intensity

Panel A: Sentiment Type Intensity in Risk Factor Disclosure

$$SENT_{RF,it} = \beta_0 + \beta_1 SIZE_{it} + \beta_2 BTM_{it} + \beta_3 VWRET_{it} + \beta_4 LEV_{it} + \beta_5 STDSRV_{it} + \beta_6 BETA_{it} + \beta_7 DNI_{it} + \beta_8 ANALYSTS_{it} + \beta_9 INSTOWN_{it} + \text{year effects}_t + \text{industry effect}_i + \varepsilon_{it} \quad (1)$$

Variables	POS_{RF}	NEG_{RF}	UNC_{RF}	LIT_{RF}	WMD_{RF}	MMD_{RF}	SMD_{RF}	CNT_{RF}	FOG_{RF}
Intercept	1.159	4.427	3.876	2.016	3.051	0.790	0.408	1.241	15.599
	6.847***	8.818***	12.535***	6.077***	10.825***	7.681***	5.173***	7.460***	10.073***
SIZE	−0.045	−0.038	−0.048	0.121	−0.081	0.015	−0.007	−0.017	0.551
	−2.502**	−0.652	−1.521	3.36***	−2.995***	1.382	−0.828	−1.068	2.109**
BTM	−0.038	0.036	0.089	0.004	−0.036	0.014	−0.001	−0.008	0.157
	−1.180	0.458	1.737*	0.089	−0.971	0.957	−0.068	−0.251	0.634
VWRET	−0.185	0.122	−0.473	0.455	0.243	0.187	−0.088	0.246	6.209
	−0.731	0.156	−0.646	0.817	0.460	1.451	−0.771	0.818	2.392**
LEV	−0.020	0.190	−0.077	0.072	−0.048	0.007	0.002	0.052	0.669
	−0.218	0.600	−0.486	0.506	−0.272	0.180	0.036	0.527	0.503
STDSRV	−0.438	−7.795	−3.106	−2.936	1.331	−1.052	2.428	−0.312	4.755
	−0.246	−1.284	−0.713	−0.816	0.407	−1.202	1.905*	−0.171	0.357
BETA	−0.087	0.055	0.133	−0.112	0.033	0.052	−0.014	0.175	−0.180
	−1.286	0.294	1.351	−1.110	0.368	1.612	−0.512	3.275***	−0.439
DNI	0.055	0.240	0.203	−0.054	0.191	0.039	0.014	0.019	0.378
	1.780*	4.213***	4.767***	−1.491	4.121***	3.564***	1.747*	0.597	1.585
ANALYSTS	0.074	0.123	0.046	−0.175	0.079	−0.054	−0.020	−0.047	−0.301
	1.778*	1.128	0.698	−2.026**	1.555	−2.560**	−1.031	−1.209	−0.765
INSTOWN	−0.147	0.022	0.004	0.391	0.075	−0.019	−0.028	0.091	−0.464
	−1.165	0.101	0.028	1.920*	0.797	−0.580	−0.671	1.254	−1.261
R^2	0.433	0.333	0.399	0.407	0.463	0.401	0.369	0.406	0.370
Adjusted R^2	0.334	0.216	0.295	0.304	0.370	0.297	0.259	0.302	0.260

*, **, *** Denote 10 percent, 5 percent, and 1 percent two-tailed statistical significance levels, respectively.
n = 378. Year and industry fixed effects included. Standard errors clustered by firm.
Appendix A contains the definitions of all variables in this table.

(continued on next page)

Given their high coefficient values, the multivariate results suggest that *LIT* and *FOG* are strong proxies for litigation risk among firms that have experienced cybersecurity incidents before their filings.

Second, as a proxy for financial risk, we found no significant associations between sentiment types and *LEV* in the RF disclosure, which suggests that sentiment types are not informative of a company's financial risk. However, *DNI*, a profitability proxy, is positively associated with NEG_{RF} (0.240, p = 0.000), UNC_{RF} (0.203, p = 0.000), WMD_{RF} (0.191, p = 0.000), and MMD_{RF} (0.039, p = 0.000). Moreover, given that DNI is calculated with net income before extraordinary items (*NI*) deflated by the firm's lagged market value of equity, these sentiment types provide a new and valuable perspective on profitability proxies.

Third, research shows that market beta (*BETA*) is a proxy for market risks and firms' other systematic risks (Campbell et al. 2014). *BETA* is positively related to CNT_{RF} (0.175, p = 0.001). The result showing a positive relation between a contingent sentiment in the risk disclosure of breach firms with market risks and firm's systematic risks is reasonable. Thus, CNT_{RF} may be a proxy for firms' other systematic risks.

For other idiosyncratic risks proxies, such as *STDSRV* and *VWRET*, we find that LIT_{RF} is positively negatively associated with *STDSRV* (2.428, p = 0.057) and FOG_{RF} is positively associated with *VWRET* (2.109, p = 0.017). One possible explanation for our results is the homogeneity of our sample population. Given our samples consist of only breached firms, the sentiment types reveal no additional meaningful insight regarding companies' unique risks.

In summary, sentiment types from risk factor disclosures are informative regarding the different risks that a firm is experiencing. For example, higher *LIT* and lower *POS* and *WMD* indicate litigation risk, whereas a higher level of

TABLE 4 (continued)

Panel B: Sentiment Type Intensity in Management Discussion and Analysis Disclosure

$$SENT_{MDA,it} = \beta_0 + \beta_1 SIZE_{it} + \beta_2 BTM_{it} + \beta_3 VWRET_{it} + \beta_4 LEV_{it} + \beta_5 STDSRV_{it} + \beta_6 BETA_{it} + \beta_7 DNI_{it} + \beta_8 ANALYSTS_{it} + \beta_9 INSTOWN_{it} + \text{year effects}_t + \text{industry effect}_i + \varepsilon_{it} \quad (1)$$

Variables	POS_{MDA}	NEG_{MDA}	UNC_{MDA}	LIT_{MDA}	WMD_{MDA}	MMD_{MDA}	SMD_{MDA}	CNT_{MDA}	FOG_{MDA}
Intercept	0.269	2.325	1.110	2.573	0.298	0.369	0.163	0.847	16.538
	1.922*	9.147***	4.693***	10.965***	3.84***	5.370***	3.493***	6.471***	9.149***
SIZE	0.045	0.024	0.023	0.003	−0.006	−0.004	0.008	−0.020	−0.032
	3.029***	0.810	1.048	0.194	−0.769	−0.625	1.433	−1.290	−0.206
BTM	0.014	0.200	0.054	−0.005	−0.009	−0.008	0.000	−0.057	0.116
	0.595	3.221***	1.011	−0.129	−0.581	−0.667	0.01	−2.107**	0.408
VWRET	−0.021	0.057	−0.113	−0.033	0.014	−0.168	0.057	−0.236	1.778
	−0.113	0.153	−0.345	−0.114	0.112	−1.208	0.777	−1.072	0.886
LEV	−0.074	0.222	−0.130	0.015	−0.040	−0.040	0.018	0.023	1.109
	−0.851	1.806*	−1.264	0.170	−1.050	−1.009	0.796	0.318	1.265
STDSRV	−1.650	0.789	−0.965	−1.814	−0.958	−0.765	0.570	−1.814	−0.288
	−1.012	0.213	−0.381	−0.810	−1.165	−0.957	0.873	−1.002	−0.013
BETA	0.079	0.144	0.242	−0.102	0.031	0.020	−0.026	0.024	1.159
	1.344	1.306	2.554**	−1.170	1.022	0.764	−1.243	0.416	2.113**
DNI	0.071	0.017	0.092	−0.011	0.005	0.037	0.013	−0.012	0.641
	4.969***	0.384	3.837***	−0.432	0.776	4.709***	1.267	−0.456	2.488**
ANALYSTS	−0.073	−0.139	0.021	−0.019	0.018	0.020	0.001	−0.006	0.048
	−2.457**	−1.546	0.427	−0.439	1.168	1.279	0.054	−0.140	0.150
INSTOWN	−0.011	0.107	−0.163	0.102	−0.013	−0.116	−0.055	−0.058	−0.511
	−0.181	0.683	−1.117	1.593	−0.331	−2.069**	−2.313**	−0.592	−0.735
R^2	0.423	0.577	0.369	0.580	0.367	0.294	0.298	0.309	0.324
Adjusted R^2	0.323	0.503	0.258	0.507	0.257	0.171	0.175	0.188	0.207

*, **, *** Denote 10 percent, 5 percent, and 1 percent two-tailed statistical significance levels, respectively.
n = 378. Year and industry fixed effects included. Standard errors clustered by firm.
Appendix A contains the definitions of all variables in this table.

CNT suggests other systematic risks of a company. Furthermore, the positive associations of *NEG*, *UNC*, and *WMD* with *DNI*, a profitability measure, postulate a new and meaningful profitability proxy. Taking together, regression results of sentiment types as risk proxies support H1.

Sentiment Types and Cumulative Abnormal Returns

We hypothesize that sentiment types are predictors of market reactions, specifically with cumulative abnormal returns (CAR). As such, we operationalize each sentiment type in the model individually to assess its unique association with CAR. For a comparative analysis of the informativeness of the content within the SEC report, we test the sentiment types by the disclosure types of the SEC report (i.e., RF and MDA). Adopting prior information content literature (Campbell et al. 2014; Han, Jennings, and Noel 1992; Wahlen 1994) and data breach research (Richardson et al. 2019), we estimate the following equation to test H2. *Type* represents risk factor or management discussion and analysis, *i* represents the firm, and *t* represents the fiscal year:

$$\begin{aligned} CAR_{Type,it} = {} & \beta_0 + \beta_1 POS_{Type,it} + \beta_2 NEG_{Type,it} + \beta_3 UNC_{Type,it} + \beta_4 LIT_{Type,it} + \beta_5 WMD_{Type,it} \\ & + \beta_6 MMD_{Type,it} + \beta_7 SMD_{Type,it} + \beta_8 CNT_{Type,it} + \beta_9 FOG_{Type,it} + \beta_{10} \Delta NI_{it} + \beta_{11} LOSSES_{it} \\ & + \beta_{12} AVGACC_{it} + \beta_{13} LEV_{it} + \beta_{14} BETA_{it} + \beta_{15} STDSRV_{it} + \beta_{16} BTM_{it} + \beta_{17} SIZE_{it} \\ & + \beta_{18} ICW_{it} + \beta_{19} RESTATE_{it} + \beta_{20} VWRET_{it} + \beta_{21} STDR_{it} + \beta_{22} TVOL_{it} + \text{ year effects}_t \\ & + \text{industry effects}_i + \varepsilon_{it} \end{aligned} \quad (2)$$

Consistent with prior risk disclosure literature (Filzen 2015; Filzen et al. 2023; Hammersley et al. 2008; Wang et al. 2013), we calculate *CAR* based on the Fama and French (1996) three-factor model as the market risk model with 100 days in our estimation window and a minimum of 70 observations, which end two days before the event window of one day before and one day after the SEC filing date.

Since short-window returns, such as *CAR*, are likely to be influenced by firms' expected cash flows and investors' perceptions of firm risks (Campbell et al. 2014; Kothari 2001; Kothari et al. 2009), we also follow prior literature to control for the information that affects these factors. We include the change in net income scaled by lagged market equity (ΔNI) and a dummy variable that depicts net losses of firms over the previous five years (*LOSSES*) to control for firms' earnings surprises, which are key factors affecting a firm's future cash flows. Additionally, the absolute value of accruals (*AVGACC*) and financial leverage (*LEV*) from Equation (1) are retained because they are likely to affect future cash flows. To address the perception of firm risks, we include the market beta (*BETA*) to control for other systematic and overall market risks, stock return volatility (*STDSRV*) and prior market returns (*VWRET*) to control for other idiosyncratic risks, and firm size (*SIZE*) to control for litigation risks. Furthermore, we include internal control weaknesses (*ICW*), financial restatement (*RESTATE*), daily stock volatility (*STDR*), and share turnover (*TVOL*) as additional control variables for other idiosyncratic risks and comparability with prior literature (Richardson et al. 2019). Consistent with prior research (Campbell et al. 2014; Richardson et al. 2019) all regression testing includes year and industry fixed effects with standard errors clustered by firms.

Overall (see Table 5), the sentiment types in both RF and MDA exhibited informativeness, as measured by *CAR* in the short-window period (−1,1). As predicted, *CAR* is negatively associated with negative sentiment (*NEG*) in RF (−0.018, p = 0.000). Based on the average of \$16.76 billion ($e^{9.727}$) in market capitalization value in our sample, changes in negative sentiment represent a multiplier of over \$300 million of reduction in a firm's valuation over the three days around the filing. We also find a significant positive relationship between *CAR* and *WMD* (0.028, p = 0.044), and estimate its economic significance at \$469 million, thus supporting H2. This finding is critical for cybersecurity risk and sentiment research and has far-reaching implications for the management of breach firms and regulators.

In the MDA section, the test finds a significant association between *LIT* and *CAR* (−0.030, p = 0.028), translating to an economic significance of \$503 million. The results support H3 that investors react differently to the informativeness of sentiment types in the different sections of the SEC report. Our results reflect a more substantial economic significance than extant risk disclosure research (Filzen 2015; Richardson et al. 2019; Campbell et al. 2014). We are also the first to provide empirical evidence that sentiment types associate with market reactions differently between RF and MDA of SEC reports. Furthermore, this study extends the application of the LM word list as a content analysis tool from the SEC report to the report section level. Together with our model focusing on the sentiment types, this extension provides fertile ground for future research in this field.

Alternative Testing Using Partial Least Squares Structural Equation Modeling

After completing this testing using OLS regression, which examines each endogenous variable individually, we applied an alternative analytical method using PLS-SEM. A foundationally regression-based statistical method, PLS-SEM is a causal predictive approach that emphasizes prediction in a structural model (Hair, Risher, Sarstedt, and Ringle 2019b; Shmueli et al. 2019). Furthermore, unlike covariance-based SEM, which focuses on the common variance to measure the goodness of fit (Sarstedt, Hair, Ringle, Thiele, and Gudergan 2016), PLS-SEM maximizes the variances extracted from the exogenous and endogenous variables in the model to explain the causal relationships (Hair et al. 2017; Hair, Howard, and Nitzl 2019a). As such, we include trading volume (*TVOL*) (Kravet and Muslu 2013) and stock price bid/ask spread (*SPREAD*) (Campbell et al. 2014; Gordon, Loeb, and Sohail 2010) as another disclosure informativeness and value relevance measure in the model to better simulate the complexity of market reactions.

To begin PLS-SEM testing, we create the structural path model using SmartPLS 3 (Ringle, Wende, and Becker 2015). The model includes independent and control variables from each hypothesis tested in one structural path. In addition, we create a dichotomous variable for each year for the year-fixed effects. The structural path model uses the initial algorithm setting, which sets a path-weighting scheme of 300 maximum iterations and a stop criterion of 7. The algorithm calculates the adjusted R^2 value for each endogenous variable in the structural path, as shown in Figure 1. Given that all the variables tested are single-item measures, all the variables' outer variance inflation factor (*VIF*) is 0. However, the inner model *VIF* revealed that *UNC* and *WMD* of the RF sentiment types had a *VIF* value of 3.927 and 4.719, respectively. The values are higher than the threshold of 3.0 but less than 5.0, indicating a possible multicollinearity issue (Hair et al. 2017; Hair et al. 2019b). The PLS-SEM algorithm calculates the adjusted R^2 for *CAR* (−1,1),

TABLE 5

Multivariate Analysis by Sentiment Type Intensity and *CAR* (−1,1)

$$\begin{aligned} CAR_{Type,it} = {} & \beta_0 + \beta_1 POS_{Type,it} + \beta_2 NEG_{Type,it} + \beta_3 UNC_{Type,it} + \beta_4 LIT_{Type,it} + \beta_5 WMD_{Type,it} + \beta_6 MMD_{Type,it} \\ & + \beta_7 SMD_{Type,it} + \beta_8 CNT_{Type,it} + \beta_9 FOG_{Type,it} + \beta_{10} \Delta NI_{it} + \beta_{11} LOSSES_{it} + \beta_{12} AVGACC_{it} \\ & + \beta_{13} LEV_{it} + \beta_{14} BETA_{it} + \beta_{15} STDSRV_{it} + \beta_{16} BTM_{it} + \beta_{17} SIZE_{it} + \beta_{18} ICW_{it} + \beta_{19} RESTATE_{it} \\ & + \beta_{20} VWRET_{it} + \beta_{21} STDR_{it} + \beta_{22} TVOL_{it} + \text{year effects}_t + \text{industry effects}_i + \varepsilon_{it} \end{aligned} \quad (2)$$

	Independent Variable Type: *RF*		Independent Variable Type: *MDA*	
Variables	**Coeff.**	**t-statistic**	**Coeff.**	**t-statistic**
Intercept	0.010	0.139	0.048	0.982
POS_{TYPE}	−0.012	−0.828	−0.021	−1.433
NEG_{TYPE}	−0.018	−3.766***	0.004	0.502
UNC_{TYPE}	−0.011	−0.963	−0.012	−1.401
LIT_{TYPE}	0.001	0.076	−0.030	−2.199**
WMD_{TYPE}	0.028	2.019**	−0.016	−0.418
MMD_{TYPE}	0.016	0.627	−0.032	−0.973
SMD_{TYPE}	−0.027	−1.036	−0.006	−0.116
CNT_{TYPE}	0.001	0.086	0.009	0.485
FOG_{TYPE}	0.001	0.413	0.001	0.886
ΔNI_{it}	−0.003	−1.865*	−0.002	−1.929*
$LOSSES_{it}$	0.004	0.416	0.002	0.253
$AVGACC_{it}$	−0.058	−0.663	−0.068	−0.809
LEV_{it}	0.014	1.022	0.003	0.212
$BETA_{it}$	0.001	0.044	0.001	0.065
$STDSRV_{it}$	−1.363	−1.606	−1.367	−1.506
BTM_{it}	0.019	1.976**	0.016	1.611
$SIZE_{it}$	−0.003	−1.147	−0.002	−0.877
ICW_{it}	−0.030	−1.636	−0.032	−2.617***
$RESTATE_{it}$	−0.022	−1.331	−0.019	−1.228
$VWRET_{it}$	0.059	0.904	0.076	1.186
$STDR_{it}$	0.299	0.748	0.329	0.818
$TVOL_{it}$	0.000	0.005	−0.001	−0.092
R^2	0.245		0.239	
Adjusted R^2	0.076		0.068	

*, **, *** Denote 10 percent, 5 percent, and 1 percent two-tailed statistical significance levels, respectively.
n = 378. Year and industry fixed effects included. Standard errors clustered by firm.
Appendix A contains the definitions of all variables in this table.

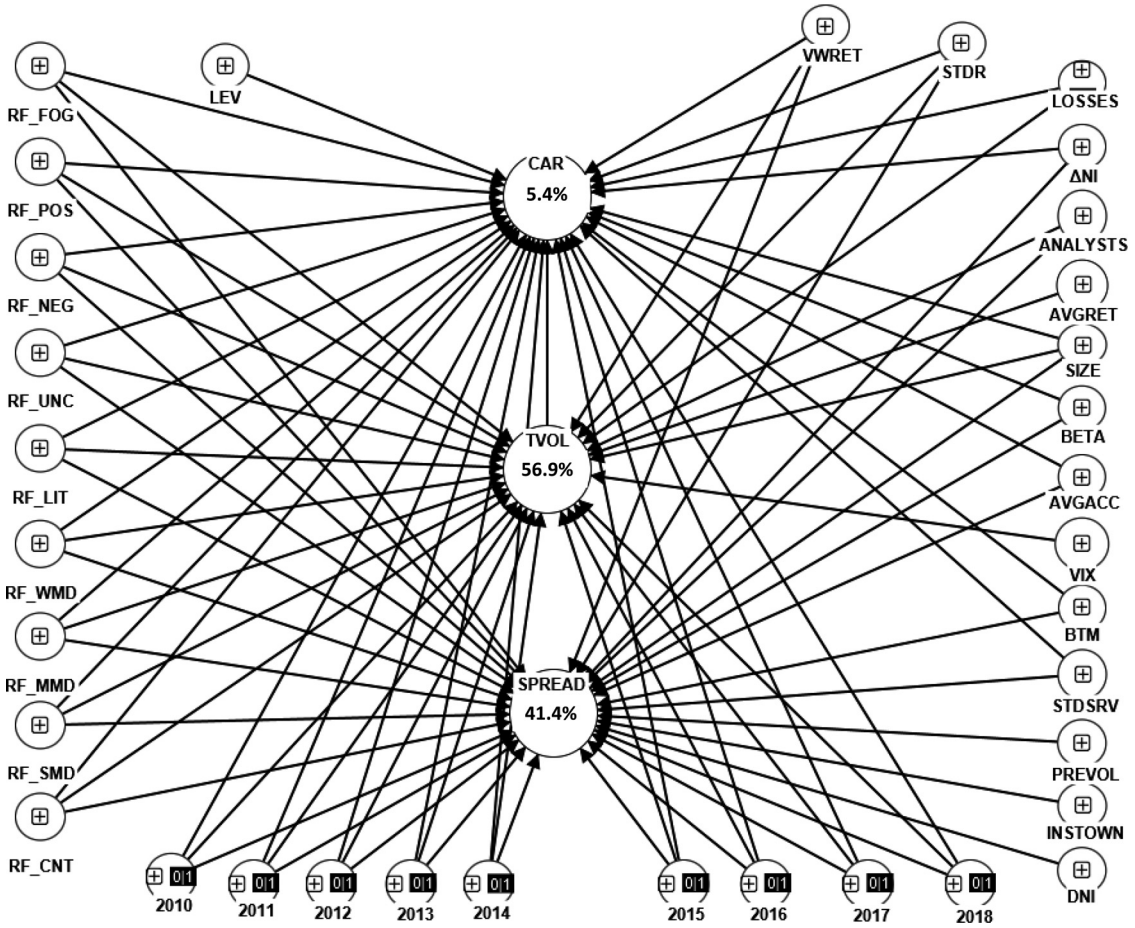
TVOL (−1,1), and *SPREAD* (−1,1) as 0.054, 0.569, and 0.414, respectively. The results are directionally similar to those calculated using OLS equations.[13]

Next, we perform the bootstrapping procedure to assess the strength of the relationships and statistical significance of each structural path. The parameters of the bootstrapping procedure include 5,000 bootstrap samples[14] and a 0.1 significance level two-tailed test using a bias-corrected and accelerated (BCa) confidence interval method (Hair et al. 2017). We apply the control variables for each endogenous variable based on prior literature. Thus, a blank appears in the

[13] Separate OLS calculations of *TVOL* and *SPREAD* using Equation (2) were made to compare the results from PLS-SEM.

[14] As a robustness test of using 10,000 subsamples for bootstrapping, the p-values for all paths are practically the same as the main analysis using 5,000 subsamples. Standard errors are not clustered by firm in PLS-SEM since the results are bootstrapped and BCa confidence interval method (Hair et al. 2017).

FIGURE 1
PLS Structural Model of RF Sentiment Types with Adjusted R^2



column of an endogenous variable means the control variable is not applicable. The results of the bootstrapping procedure in Table 9 are consistent with the OLS results presented earlier in the hypothesis testing.

SmartPLS allows us to test *CAR*, *TVOL*, and *SPREAD* simultaneously in a path model with different combinations of explanatory variables. As such, we model *CAR* testing as shown in Equation (2) except for *ICW*, *RESTATE*, and industry fixed effect, as these variables contain insufficient variability (i.e., primarily zeros). We also follow previous research (Kravet and Muslu 2013) to test *TVOL* and include additional control variables: average stock returns (*AVGRET*), average trading volume for the 60-day trading period ending two days before the estimation period (*PREVOL*), a market stock volatility index (*VIX*), and the number of analysts following a firm (*ANALYSTS*) as control variables. Similarly, while keeping most of the control variables from the CAR model, we include institutional ownership (*INSTOWN*) and net income before extraordinary items (*DNI*) as control variables for *SPREAD*, which is consistent with the literature (Campbell et al. 2014). Again, the results are consistent with prior research; however, the PLS-SEM results of RF and MDA disclosures reveal additional meaningful insights into the sentiment types.

Specifically, Table 6 shows that *CAR* associates negatively with *NEG* (−0.012, p = 0.007) but positively with *WMD* (0.025, p = 0.061). The coefficients and levels of statistical significance of these two associations between OLS and PLS are directionally similar. *TVOL* is associated negatively with *POS* (−0.792, p = 0.007), *UNC* (−0.498, p = 0.025), *LIT* (−0.095, p = 0.026), and *CNT* (−0.525, p = 0.066) but positively with *SMD* (1.020, p = 0.045). *SPREAD* is negatively associated with *POS* (−0.051, p = 0.078). As such, the results of PLS-SEM reaffirm the main testing supporting H2 and provide additional insight that sentiment types are informative in the presence of multiple endogenous variables that represent a more realistic simulation of the actual market environment.

TABLE 6

PLS-SEM Structural Path Coefficients and t-statistics of RF Sentiment Types for *CAR*, *TVOL*, and *SPREAD*

	CAR (−1,1)		*TVOL* (−1,1)		*SPREAD* (−1,1)	
	Coeff.	**t-statistic**	**Coeff.**	**t-statistic**	**Coeff.**	**t-statistic**
POS_{RF}	−0.014	1.256	−0.792	2.701***	−0.051	1.762*
NEG_{RF}	−0.012	2.687***	0.043	0.393	−0.004	0.535
UNC_{RF}	−0.015	1.482	−0.498	2.236**	−0.011	0.609
LIT_{RF}	−0.005	0.745	−0.095	0.636	−0.014	1.285
WMD_{RF}	0.025	1.871*	0.631	2.227**	0.004	0.159
MMD_{RF}	−0.005	0.218	−0.160	0.262	0.142	1.566
SMD_{RF}	−0.031	1.226	1.020	2.008**	−0.031	0.727
CNT_{RF}	0.002	0.118	−0.525	1.840*	−0.005	0.171
FOG_{RF}	0.001	0.511	0.004	0.142	−0.001	0.172
ΔNI_{it}	−0.002	0.672			0.000	0.002
$ANALYSTS_{it}$			0.726	5.372***	−0.027	1.231
$AVGACC_{it}$	−0.032	0.523			−0.288	0.929
$AVGRET_{it}$			0.184	1.686*		
$BETA_{it}$	−0.007	0.563			−0.060	1.162
BTM_{it}	0.016	1.725*			0.040	3.121***
DNI_{it}					0.023	1.096
$INSTOWN_{it}$					−0.124	3.554***
LEV_{it}	−0.001	0.073				
$LOSSES_{it}$	0.005	0.612	0.205	1.188		
$PREVOL_{it}$					−0.053	1.557
$SIZE_{it}$	−0.002	1.022	−0.289	4.122***	−0.025	2.570***
$STDR_{it}$	0.228	0.656	41.349	9.795***	0.290	1.050
$STDSRV_{it}$	−0.930	1.393			4.074	2.791***
$TVOL_{it}$	−0.001	0.273				
VIX_{it}			−0.010	0.568		
$VWRET_{it}$	0.043	0.651	−0.346	0.240	−0.068	0.591
Adjusted R^2	0.054		0.569		0.414	

*, **, *** Denote 10 percent, 5 percent, and 1 percent two-tailed statistical significance levels, respectively.
n = 378. Year fixed effects are included in the model and operationalized as dummy variables.
Appendix A contains the definitions of all variables in this table.

We also apply the same PLS-SEM path model shown in Figure 1 to MDA sentiment types, and Table 7 contains the results. First, in addition to a significant negative association with *UNC* (−0.017, p = 0.028) and *LIT* (−0.014, p = 0.058), *CAR* is also positively associated with *FOG* (0.002, p = 0.100). The path model yields a 5.4 percent adjusted R^2 for *CAR*, the same as in the RF sentiment types. Second, the results from the PLS-SEM testing between RF and MDA sentiment types reveal significant associations of different sentiment types, supporting the OLS results for H2. This finding also suggests that investors perceive risks through distinctly different sentiment types between RF and MDA, thus supporting H3. The results definitively answer the first research question that sentiment types are informative at the report level and at the subsection level of a report, such as RF and MDA. In short, our empirical results provide fertile ground for future research on disclosure and sentiment.

In MDA sentiment types, *TVOL* is negatively associated with *POS* (−0.562, p = 0.021) and *LIT* (−0.404, p = 0.050) but positively with *SMD* (1.276, p = 0.099). Although only three sentiment types are significantly associated with *TVOL* in MDA disclosure, the results suggest that RF and MDA sentiment types are predictors of trading volatility in the short window (−1.1), with a high predictive value of 56.9 percent adjusted R^2 in both RF and MDA models. The *TVOL* results affirm the answer to our research question that sentiment types are informative to the subsection level of SEC reports and support H3 that different sentiment types are relevant to investors differently between RF and MDA disclosures.

TABLE 7

PLS-SEM Structural Path Coefficients and t-statistics of MDA Sentiment Types for *CAR*, *TVOL*, and *SPREAD*

	***CAR* (−1,1)**		***TVOL* (1,1)**		***SPREAD* (−1,1)**	
	Coeff.	**t-statistic**	**Coeff.**	**t-statistic**	**Coeff.**	**t-statistic**
POS_{RF}	−0.012	1.091	−0.562	2.304**	−0.026	0.895
NEG_{RF}	0.004	0.590	0.150	0.893	0.027	1.189
UNC_{RF}	−0.017	2.191**	−0.204	1.324	0.000	0.011
LIT_{RF}	−0.014	1.897*	−0.404	1.956**	−0.035	1.433
WMD_{RF}	0.018	0.596	0.754	0.903	0.009	0.142
MMD_{RF}	−0.037	1.192	0.114	0.143	−0.121	1.208
SMD_{RF}	−0.041	1.076	1.276	1.650*	0.126	1.808*
CNT_{RF}	0.007	0.448	−0.103	0.310	0.017	0.518
FOG_{RF}	0.002	1.645*	−0.059	1.460	−0.001	0.447
ΔNI_{it}	−0.003	0.703			0.000	0.034
$ANALYSTS_{it}$			0.679	4.809***	−0.030	1.268
$AVGACC_{it}$	−0.044	0.684			−0.299	0.969
$AVGRET_{it}$			0.218	1.890*		
$BETA_{it}$	−0.008	0.620			−0.042	0.911
BTM_{it}	0.016	1.680*			0.041	2.809***
DNI_{it}					0.024	1.210
$INSTOWN_{it}$					−0.125	3.643***
LEV_{it}	−0.009	0.510				
$LOSSES_{it}$	0.003	0.380	0.212	1.135		
$PREVOL_{it}$					−0.061	1.615
$SIZE_{it}$	−0.001	0.658	−0.284	3.794***	−0.021	2.343**
$STDR_{it}$	0.253	0.733	40.693	9.894***	0.283	1.025
$STDSRV_{it}$	−0.917	1.366			3.896	2.751***
$TVOL_{it}$	−0.002	0.329				
VIX_{it}			−0.008	0.427		
$VWRET_{it}$	0.050	0.814	−0.038	0.071		0.447
Adjusted R^2	0.054		0.569		0.410	

*, **, *** Denote 10 percent, 5 percent, and 1 percent two-tailed statistical significance levels, respectively.
n = 378. Year fixed effects are included in the model and operationalized as dummy variables.
Appendix A contains the definitions of all variables in this table.

Like the RF sentiment testing, we only find one significant association between the MDA sentiment type and *SPREAD* (*SMD* 0.126, p = 0.071) with a 41.0 percent adjusted R^2. Although the model has a high level of predictability, the lack of significant associations with *SPREAD* among the sentiment types in both RF and MDA disclosure suggests that sentiment types are not contributing meaningfully to the predictability of *SPREAD*. Table 7 also shows the adjusted R^2 results (i.e., *CAR* = 0.050, *TVOL* = 0.477, and *SPREAD* = 0.248) from MDA testing comparable to those in RF. Again, the adjusted R^2 results from PLS-SEM were the same as those in the OLS calculations supporting the validity of the test results.

Sentiment Types Predictive Power

SmartPLS 3 (Ringle, Wende, and Becker 2015) incorporates a function called PLSpredict[15] ($Q^2_{predict}$) to assess the out-of-sample predictive power of a structural model (Shmueli et al. 2019). We used the recommended parameter of 10 folds and 10 repetitions for PLSpredict. The results for RF sentiment types show positive $Q^2_{predict}$ values for *SPREAD* (0.262) and *TVOL* (0.513), which suggests that the PLS model has an out-of-sample predictive power for *SPREAD* and

[15] See Shmueli et al. (2019) for the methodological discussion on PLSpredict.

TVOL. Likewise, the PLSpredict results for MDA sentiment types also demonstrate positive $Q^2_{predict}$ values for *SPREAD* (0.234) and *TVOL* (0.520). However, the $Q^2_{predict}$ values of *CAR* under both RF and MDA sentiment types are less than 0, which indicates a lack of out-of-sample predictive power. Although we follow prior literature in the equation estimation for *CAR*, including control variables (Campbell et al. 2014; Han et al. 1992; Wahlen 1994), only *BTM* is significant. One possible explanation is that the model does not capture more relevant control variables. New control variables could be a future research opportunity to improve the predictive relevance of Equation (1) for the CAR.

Nevertheless, given the causal-predictive statistical estimation of PLS-SEM, the results of the bootstrapping procedure extend the association relationships found in our regression tests to a causal relationship. Specifically, among the sentiment types in RF disclosure, empirical evidence supports that *NEG* and *WMD* are predictors of CAR; *POS*, *UNC*, *WMD*, *SME*, and *CNT* are predictors of *TVOL*. Additionally, *POS* is the predictor of *SPREAD*. Furthermore, in MDA sentiment types, *UNC* and *LIT* are predictors of CAR; *POS*, *LIT*, and *SMD* are predictors of *TVOL*, whereas *SMD* is the predictor of *SPREAD*. The PLSpredict results also support the out-of-sample predictive models for *TVOL* and *SPREAD*.

Robustness Tests

Our sample contains all incidents reported in the PRC from 2009 to 2018. Inevitably, companies may experience multiple incidents in the same year. To test the robustness of the model and the results earlier, we have removed 136 samples representing multiple incidents of the same company in the same year or quarter. Next, we test our H2 on RF and MDA sentiment types with the remaining non-repeating incidents of 242, and the results are shown in Tables 8 and 9, respectively.

Table 8 compares the results of the RF sentiments from the original sample set of 378 to the non-repeating sample of 242. The *NEG* sentiment continues to show a significant association with *CAR* (−0.024, p = 0.001), with an increase of 6 basis points higher than the original sample. In contrast, the association between *WMD* and *CAR* becomes nonsignificant. Additionally, the adjusted R^2 of the nonrepeating sample set has improved from 0.076 to 0.136, suggesting more explanatory power of H2 in the nonrepeating sample set compared to the original sample set.

Table 9 compares the MDA sentiments between the original sample set and the nonrepeating sample set. The results in the nonrepeating samples suggest that not only *LIT*'s association with *CAR* is significant but also stronger at −0.041 (p = 0.043) as compared to the original sample set (LIT_{MDA} − 0.030, p = 0.028). We also find other significant associations in MDA sentiment with *CAR* in *POS* (0.033, p = 0.037) and *UNC* (−0.028, p = 0.013). Similar to the improved results in RF, the adjusted R^2 of the nonrepeating sample of MDA also improves from 0.068 to 0.110.

We also analyze the differences in sentiment types among our samples pre- and post-SEC 2011 guidance. An independent-samples test reveals no significance across all sentiment types except *MMD* in RF (p = 0.029) and *NEG* in MDA (p = 0.000). Combining the results of RF and MDA in this robustness test, sentiment types maintain their significant associations with *CAR*, and the overall explanatory capacity, as reflected in the adjusted R^2 has also improved, supporting H2 and H3.

Sensitivity Tests

As mentioned earlier, the paucity of studies about the relationship between the full spectrum of sentiment types and market reactions restricts our understanding of their relationships and motivates this study. Recent studies have performed match sampling to compare the difference in the litigious tone in cybersecurity disclosure (Gao et al. 2020) and short-window CAR (Richardson et al. 2019), confirming statistical significance between breached and nonbreached firms. For comparability and causality analysis, we adopt the genetic matching (GM) algorithm[16] (Diamond and Sekhon 2013). This multivariate matching method uses an algorithm to determine each covariate's weight in selecting a nonbreached subsample of 133 from the unique breached firm observations. An independent-samples test on the sentiment types in RF and MDA between the breached and matched nonbreach firms confirms that the disclosure sentiment between the two groups differs. Overall, our results show that readability (*FOG*) scores for breach firms are significantly different[17] than the nonbreach firms (p < 0.001) in both RF and MDA disclosures. More specifically, the t-statistic shows that in RF disclosure, *UNC* significantly differs (p < 0.022) between breached and nonbreached observations in the GM sample.

On the other hand, we find *POS* (p = 0.100), *MMD* (p = 0.074), and *CNT* (p = 0.100) are significantly different in the MDA disclosure between the genetically matched groups. The t-statistic results from the genetic matching affirm our

[16] See Appendix C for the genetic matching algorithm processing and balance analysis.
[17] One-tailed t-statistic. Consistent with Richardson et al. (2019).

TABLE 8

Robustness Test of RF Sentiment Type Intensity and *CAR* (−1,1)

$$\begin{aligned} CAR_{Type,it} = {} & \beta_0 + \beta_1 POS_{Type,it} + \beta_2 NEG_{Type,it} + \beta_3 UNC_{Type,it} + \beta_4 LIT_{Type,it} + \beta_5 WMD_{Type,it} + \beta_6 MMD_{Type,it} \\ & + \beta_7 SMD_{Type,it} + \beta_8 CNT_{Type,it} + \beta_9 FOG_{Type,it} + \beta_{10} \Delta NI_{it} + \beta_{11} LOSSES_{it} + \beta_{12} AVGACC_{it} \\ & + \beta_{13} LEV_{it} + \beta_{14} BETA_{it} + \beta_{15} STDSRV_{it} + \beta_{16} BTM_{it} + \beta_{17} SIZE_{it} + \beta_{18} ICW_{it} + \beta_{19} RESTATE_{it} \\ & + \beta_{20} VWRET_{it} + \beta_{21} STDR_{it} + \beta_{22} TVOL_{it} + \text{year effects}_t + \text{industry effects}_i + \varepsilon_{it} \end{aligned} \quad (2)$$

	Multiple Incidents Company Year		Single Incident Company Year	
Variables	**Coeff.**	**t-statistic**	**Coeff.**	**t-statistic**
Intercept	0.010	0.139	0.038	0.533
POS_{RF}	−0.012	−0.828	0.000	−0.014
NEG_{RF}	−0.018	−3.766***	−0.024	−3.231***
UNC_{RF}	−0.011	−0.963	−0.017	−1.015
LIT_{RF}	0.001	0.076	0.003	0.248
WMD_{RF}	0.028	2.019**	0.034	1.607
MMD_{RF}	0.016	0.627	0.004	0.121
SMD_{RF}	−0.027	−1.036	−0.006	−0.156
CNT_{RF}	0.001	0.086	−0.024	−1.084
FOG_{RF}	0.001	0.413	0.000	−0.179
ΔNI_{it}	−0.003	−1.865*	−0.032	−3.359***
$LOSSES_{it}$	0.004	0.416	0.009	0.700
$AVGACC_{it}$	−0.058	−0.663	−0.079	−0.653
LEV_{it}	0.014	1.022	0.005	0.320
$BETA_{it}$	0.001	0.044	0.005	0.267
$STDSRV_{it}$	−1.363	−1.606	−1.287	−1.419
BTM_{it}	0.019	1.976**	0.017	1.867*
$SIZE_{it}$	−0.003	−1.147	0.001	0.313
ICW_{it}	−0.030	−1.636	−0.026	−1.038
$RESTATE_{it}$	−0.022	−1.331	−0.017	−0.816
$VWRET_{it}$	0.059	0.904	0.026	0.321
$STDR_{it}$	0.299	0.748	0.793	1.819*
$TVOL_{it}$	0.000	0.005	−0.005	−0.685
n	378		242	
R^2	0.245		0.383	
Adjusted R^2	0.076		0.136	

*, **, *** Denote 10 percent, 5 percent, and 1 percent two-tailed statistical significance levels, respectively.
Year and industry fixed effects included. Standard errors clustered by firm.
Appendix A contains the definitions of all variables in this table.

hypotheses' tests that sentiment types other than positive and negative are informative and relevant to market reactions, thus supporting H2. Furthermore, the differences in the t-statistic further support our H3 results that investors react to different sentiment types in different sections of the SEC reports. We further analyze the durability of the differences between the genetic match groups in the longevity test next.

Sentiment Longevity Test

Following Amir et al. (2018) and Richardson et al. (2019), we extend our measurement windows of the sentiment type effect on *CAR* to observe the efficacy of the sentiment types using the nonrepeating sample of 242. Our short observation windows extend from one trading day prior and one day after the report day (−1,1) to three trading days after (−1,3) and five trading days after (−1,5). For longer observation windows, Tables 11 and 12 show the results for 10

TABLE 9

Robustness Test of MDA Sentiment Type Intensity and *CAR* (−1,1)

$$\begin{aligned} CAR_{Type,it} = {} & \beta_0 + \beta_1 POS_{Type,it} + \beta_2 NEG_{Type,it} + \beta_3 UNC_{Type,it} + \beta_4 LIT_{Type,it} + \beta_5 WMD_{Type,it} + \beta_6 MMD_{Type,it} \\ & + \beta_7 SMD_{Type,it} + \beta_8 CNT_{Type,it} + \beta_9 FOG_{Type,it} + \beta_{10} \Delta NI_{it} + \beta_{11} LOSSES_{it} + \beta_{12} AVGACC_{it} \\ & + \beta_{13} LEV_{it} + \beta_{14} BETA_{it} + \beta_{15} STDSRV_{it} + \beta_{16} BTM_{it} + \beta_{17} SIZE_{it} + \beta_{18} ICW_{it} + \beta_{19} RESTATE_{it} \\ & + \beta_{20} VWRET_{it} + \beta_{21} STDR_{it} + \beta_{22} TVOL_{it} + \text{year effects}_t + \text{industry effects}_i + \varepsilon_{it} \end{aligned} \quad (2)$$

	Multiple Incidents Company Year		Single Incident Company Year	
Variables	**Coeff.**	**t-statistic**	**Coeff.**	**t-statistic**
Intercept	0.048	0.982	0.022	0.331
POS_{MDA}	−0.021	−1.433	−0.033	−2.090**
NEG_{MDA}	0.004	0.502	0.009	0.828
UNC_{MDA}	−0.012	−1.401	−0.028	−2.492**
LIT_{MDA}	−0.030	−2.199**	−0.041	−2.027**
WMD_{MDA}	−0.016	−0.418	0.034	0.788
MMD_{MDA}	−0.032	−0.973	−0.036	−0.634
SMD_{MDA}	−0.006	−0.116	−0.002	−0.041
CNT_{MDA}	0.009	0.485	0.001	0.063
FOG_{MDA}	0.001	0.886	0.002	1.128
ΔNI_{it}	−0.002	−1.929*	−0.034	−2.174**
$LOSSES_{it}$	0.002	0.253	0.007	0.550
$AVGACC_{it}$	−0.068	−0.809	−0.098	−0.901
LEV_{it}	0.003	0.212	−0.009	−0.464
$BETA_{it}$	0.001	0.065	−0.003	−0.130
$STDSRV_{it}$	−1.367	−1.506	−0.926	−0.983
BTM_{it}	0.016	1.611	0.014	1.386
$SIZE_{it}$	−0.002	−0.877	0.002	0.701
ICW_{it}	−0.032	−2.617***	−0.024	−1.415
$RESTATE_{it}$	−0.019	−1.228	−0.014	−0.794
$VWRET_{it}$	0.076	1.186	0.038	0.450
$STDR_{it}$	0.329	0.818	0.725	1.624
$TVOL_{it}$	−0.001	−0.092	−0.005	−0.762
n	378		242	
R^2	0.239		0.365	
Adjusted R^2	0.068		0.110	

*, **, *** Denote 10 percent, 5 percent, and 1 percent two-tailed statistical significance levels, respectively.
Year and industry fixed effects included. Standard errors clustered by firm.
Appendix A contains the definitions of all variables in this table.

trading days after (−1,10), 30 trading days after (−1,30), 40 trading days after (−1,40), and 60 trading days after (−1,60) for RF and MDA sentiment, respectively.

Results in Table 10, Panel A confirm the efficacy of the RF sentiment type *NEG* with *CAR* in three days after (−0.020, p = 0.013) and five days after (−0.024, p = 0.006) with a lower adjusted R^2 after three days (5.3 percent) but higher after five days at 10.9 percent. Similarly, the efficacy of MDA sentiment types persists in the short-window observation shown in Table 10, Panel B. *POS* maintains its association with *CAR* after three days (−0.034, p = 0.054) and five days (−0.030, p = 0.081). *UNC* is still negatively associated with *CAR* after three days (−0.030, p = 0.036) and five days (−0.028, p = 0.061). Like RF, the adjusted R^2 of the short window model for MDA reduces from 11.0 percent (−1,1) to 3.6 percent after three days but increases to 8.1 percent after five days.

TABLE 10

Short Window Robustness Tests by Disclosure Type

Panel A: RF Sentiment Type Intensity and *CAR*

$$\begin{aligned} CAR_{Type,it} = {} & \beta_0 + \beta_1 POS_{Type,it} + \beta_2 NEG_{Type,it} + \beta_3 UNC_{Type,it} + \beta_4 LIT_{Type,it} + \beta_5 WMD_{Type,it} + \beta_6 MMD_{Type,it} \\ & + \beta_7 SMD_{Type,it} + \beta_8 CNT_{Type,it} + \beta_9 FOG_{Type,it} + \beta_{10} \Delta NI_{it} + \beta_{11} LOSSES_{it} + \beta_{12} AVGACC_{it} \\ & + \beta_{13} LEV_{it} + \beta_{14} BETA_{it} + \beta_{15} STDSRV_{it} + \beta_{16} BTM_{it} + \beta_{17} SIZE_{it} + \beta_{18} ICW_{it} + \beta_{19} RESTATE_{it} \\ & + \beta_{20} VWRET_{it} + \beta_{21} STDR_{it} + \beta_{22} TVOL_{it} + \text{year effects}_t + \text{industry effects}_i + \varepsilon_{it} \end{aligned} \quad (2)$$

	***CAR* (−1,3)**		***CAR* (−1,5)**	
Variables	**Coeff.**	**t-statistic**	**Coeff.**	**t-statistic**
Intercept	0.049	0.572	0.106	1.195
POS_{RF}	0.016	0.837	0.011	0.556
NEG_{RF}	−0.020	−2.474**	−0.024	−2.755***
UNC_{RF}	−0.008	−0.412	−0.013	−0.608
LIT_{RF}	−0.001	−0.067	0.000	0.008
WMD_{RF}	0.018	0.761	0.024	1.005
MMD_{RF}	−0.004	−0.100	−0.007	−0.138
SMD_{RF}	−0.025	−0.583	−0.010	−0.229
CNT_{RF}	−0.015	−0.636	−0.014	−0.562
FOG_{RF}	0.000	−0.057	−0.002	−0.674
ΔNI_{it}	−0.024	−2.036**	−0.025	−2.311**
$LOSSES_{it}$	0.008	0.491	0.001	0.051
$AVGACC_{it}$	−0.057	−0.406	−0.046	−0.310
LEV_{it}	0.019	1.006	0.027	1.319
$BETA_{it}$	−0.002	−0.072	0.010	0.382
$STDSRV_{it}$	−1.751	−1.587	−1.999	−1.718*
BTM_{it}	0.033	3.139***	0.052	4.391***
$SIZE_{it}$	0.000	−0.032	0.001	0.278
ICW_{it}	−0.028	−1.055	−0.026	−0.959
$RESTATE_{it}$	−0.005	−0.232	−0.014	−0.631
$VWRET_{it}$	0.020	0.202	0.094	0.868
$STDR_{it}$	0.734	1.179	0.795	0.932
$TVOL_{it}$	0.004	0.400	0.005	0.533
n	242		242	
R^2	0.324		0.364	
Adjusted R^2	0.053		0.109	

*, **, *** Denote 10 percent, 5 percent, and 1 percent two-tailed statistical significance levels, respectively.
Year and industry fixed effects included. Standard errors clustered by firm.
Appendix A contains the definitions of all variables in this table.

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Table 11, our longevity analysis of extended observation windows of RF sentiment types, shows that only *NEG* is significantly associated with *CAR* after 10 days (−0.022, p = 0.037) but not beyond. However, Table 12 shows a different inclination from the MDA sentiment types. Although *POS* is no longer significantly associated with *CAR* in the long windows, *LIT*'s association emerges with *CAR* 30 days after (−0.090, p = 0.031), 40 days after (−0.104, p = 0.023), and 60 days after (−0.110, p = 0.090). Like *POS*, *UNC*'s association with *CAR* becomes insignificant in the longer windows until 60 days after (−0.101, p = 0.072). Additionally, the adjusted R^2 of the model for MDA sentiment types has also improved from 3.4 percent in the −1,10 window to 16.1 percent in the −1,60 window, suggesting that the model in H2 has a more substantial explanatory attribute with longer measurement observation windows.

These longevity findings offer new insight into the efficacy of the informativeness of the sentiment types, which is different from the extant research (Richardson et al. 2019). To test the robustness of our model, we apply the longevity

TABLE 10 (continued)

Panel B: MDA Sentiment Type Intensity and *CAR*

$$CAR_{Type,it} = \beta_0 + \beta_1 POS_{Type,it} + \beta_2 NEG_{Type,it} + \beta_3 UNC_{Type,it} + \beta_4 LIT_{Type,it} + \beta_5 WMD_{Type,it} + \beta_6 MMD_{Type,it} + \beta_7 SMD_{Type,it} + \beta_8 CNT_{Type,it} + \beta_9 FOG_{Type,it} + \beta_{10}\Delta NI_{it} + \beta_{11} LOSSES_{it} + \beta_{12} AVGACC_{it} + \beta_{13} LEV_{it} + \beta_{14} BETA_{it} + \beta_{15} STDSRV_{it} + \beta_{16} BTM_{it} + \beta_{17} SIZE_{it} + \beta_{18} ICW_{it} + \beta_{19} RESTATE_{it} + \beta_{20} VWRET_{it} + \beta_{21} STDR_{it} + \beta_{22} TVOL_{it} + \text{year effects}_t + \text{industry effects}_i + \varepsilon_{it} \quad (2)$$

	***CAR* (−1,3)**		***CAR* (−1,5)**	
Variables	**Coeff.**	**t-statistic**	**Coeff.**	**t-statistic**
Intercept	0.067	0.856	0.107	1.330
POS_{MDA}	−0.034	−1.923*	−0.030	−1.742*
NEG_{MDA}	0.008	0.578	0.003	0.223
UNC_{MDA}	−0.030	−2.099**	−0.028	−1.874*
LIT_{MDA}	−0.041	−1.588	−0.045	−1.620
WMD_{MDA}	0.038	0.765	0.029	0.559
MMD_{MDA}	−0.024	−0.380	−0.032	−0.449
SMD_{MDA}	0.019	0.361	−0.019	−0.314
CNT_{MDA}	−0.003	−0.139	0.009	0.362
FOG_{MDA}	0.001	0.570	0.001	0.493
ΔNI_{it}	−0.024	−1.576	−0.028	−2.107**
$LOSSES_{it}$	0.005	0.370	−0.001	−0.085
$AVGACC_{it}$	−0.058	−0.455	−0.063	−0.463
LEV_{it}	0.006	0.274	0.016	0.705
$BETA_{it}$	−0.006	−0.233	0.002	0.093
$STDSRV_{it}$	−1.489	−1.332	−1.575	−1.315
BTM_{it}	0.030	2.609***	0.050	3.812***
$SIZE_{it}$	0.001	0.205	0.002	0.455
ICW_{it}	−0.033	−1.796*	−0.030	−1.423
$RESTATE_{it}$	−0.004	−0.223	−0.011	−0.577
$VWRET_{it}$	0.033	0.310	0.106	0.923
$STDR_{it}$	0.700	1.104	0.675	0.752
$TVOL_{it}$	0.002	0.282	0.005	0.516
n	242		242	
R^2	0.311		0.344	
Adjusted R^2	0.036		0.081	

*, **, *** Denote 10 percent, 5 percent, and 1 percent two-tailed statistical significance levels, respectively.
Year and industry fixed effects included. Standard errors clustered by firm.
Appendix A contains the definitions of all variables in this table.

test to the breached and GM nonbreached samples. Following Richardson et al. (2019), we use the OLS model as stated in H2 with a new dichotomous variable, *BREACH*, where the 133 samples from the GM receive a 0 and the breached observations receive a value of 1. The results display the statistical significance of RF and MDA sentiment types in many short and long observation windows in Tables 13 and 14, respectively.

The analysis of the RF sentiment types shows that *BREACH*, the indicator variable for the breached observations, is significantly associated with *CAR* in all windows tested, except for the five days after (−1,5) the SEC filing date. The regression coefficients and statistical significances improve from the short windows −1,1 (0.012, p = 0.086) and −1,3 (0.016, p = 0.064) to longer windows −1,10 (0.022, p = 0.038), −10,30 (0.054, p = 0.004), −1,40 (0.067, p = 0.005), and −1,60 (0.099, p = 0.019), suggesting that breach firms experience significantly more positive *CAR* as the observation

TABLE 11

Regression Results of H2 in Extended Observation Windows of RF Sentiment Types

	CAR (−1,10)		*CAR* (−1,30)		*CAR* (−1,40)		*CAR* (−1,60)	
Variables	**Coeff.**	**t-statistic**	**Coeff.**	**t-statistic**	**Coeff.**	**t-statistic**	**Coeff.**	**t-statistic**
Intercept	0.076	0.770	0.096	0.609	0.202	1.085	0.172	0.614
POS_{RF}	0.000	0.012	0.006	0.106	0.047	0.745	0.052	0.601
NEG_{RF}	−0.022	−2.089**	−0.024	−1.461	−0.020	−1.163	−0.028	−1.152
UNC_{RF}	−0.002	−0.082	0.043	1.100	0.061	1.399	0.052	0.886
LIT_{RF}	0.002	0.126	−0.008	−0.305	−0.017	−0.635	0.014	0.387
WMD_{RF}	0.014	0.476	−0.006	−0.154	−0.041	−0.870	−0.021	−0.345
MMD_{RF}	0.016	0.268	0.010	0.112	−0.023	−0.227	−0.119	−1.004
SMD_{RF}	0.000	−0.006	−0.024	−0.288	−0.050	−0.513	−0.075	−0.729
CNT_{RF}	−0.025	−0.878	−0.032	−0.778	−0.054	−1.252	−0.094	−1.602
FOG_{RF}	−0.002	−0.838	−0.001	−0.113	−0.003	−0.495	−0.001	−0.068
ΔNI_{it}	0.000	0.000	0.007	0.242	0.009	0.288	0.002	0.037
$LOSSES_{it}$	−0.006	−0.303	−0.010	−0.379	0.014	0.437	−0.004	−0.115
$AVGACC_{it}$	−0.039	−0.227	0.108	0.367	0.100	0.313	0.476	0.960
LEV_{it}	0.038	1.444	0.081	2.129**	0.098	2.480**	0.118	2.416**
$BETA_{it}$	−0.012	−0.390	−0.055	−1.071	−0.052	−0.817	−0.041	−0.468
$STDSRV_{it}$	−0.932	−0.695	0.615	0.272	1.434	0.517	1.236	0.340
BTM_{it}	0.050	3.747***	0.065	2.676***	0.076	2.996***	0.119	3.510***
$SIZE_{it}$	0.001	0.121	0.001	0.089	−0.001	−0.127	−0.007	−0.595
ICW_{it}	−0.009	−0.234	−0.045	−0.663	−0.030	−0.416	0.026	0.268
$RESTATE_{it}$	−0.026	−0.779	−0.039	−0.625	−0.053	−0.782	−0.084	−0.829
$VWRET_{it}$	0.161	1.245	0.069	0.315	−0.132	−0.492	0.022	0.059
$STDR_{it}$	0.487	0.359	−1.932	−0.614	−4.127	−1.290	−4.649	−1.138
$TVOL_{it}$	0.013	1.075	0.050	2.642***	0.063	2.971***	0.085	2.696***
Adjusted R^2		0.029		0.066		0.143		0.138

*, **, *** Denote 10 percent, 5 percent, and 1 percent two-tailed statistical significance levels, respectively.
n = 242. Year and industry fixed effects included. Standard errors clustered by firm.
Appendix A contains the definitions of all variables in this table.

windows grow longer. This predictability of the overall model (i.e., adjusted R^2) using RF sentiment types increases from 15.5 percent in the −1,1 window to 28.3 percent in the −1,60 window showing the explanatory strength and the efficacy of the model tested.

The MDA sentiment types analysis in Table 14 reveals a similar predictive longevity characteristic as in the RF sentiment types analysis. *BREACH* is not significantly associated with *CAR* in the short windows but exhibits significance starting in the long windows −1,30 (0.047, p = 0.035), −1,40 (0.061, p = 0.024), and −1,60 (0.088, p = 0.047). The predictive value of the MDA sentiment types in the GM samples resembles the results of the nonrepeating samples. The adjusted R^2 improves from 22.0 percent in the −1,30 window to 28.0 percent in the −1,60 window. The results derived from the GM samples validate those from the breached nonrepeating samples shown in Tables 11 and 12.

Taking the testing of the extended observation windows among the nonrepeating sample of 242 and the GM samples together, sentiment types beyond *POS* and *NEG* exhibit significant associations with the market reaction even after 60 days of the SEC report. The different sentiment types between RF and MDA show varying levels of longevity efficacy, with increasing values of adjusted R^2 indicating stronger predictability in longer observation windows. Our empirical results offer more explanation between disclosure and market reactions complementing extant research on disclosure sentiment and market reactions (Richardson et al. 2019; Gao et al. 2020). By incorporating all eight sentiment types, our model with an increasing adjusted R^2 in extended observation windows provides empirical support for sentiment types' importance beyond positive and negative.

TABLE 12

Regression Results of H2 in Extended Observation Windows of MDA Sentiment Types

	CAR (−1,10)		*CAR* (−1,30)		*CAR* (−1,40)		*CAR* (−1,60)	
Variables	**Coeff.**	**t-statistic**	**Coeff.**	**t-statistic**	**Coeff.**	**t-statistic**	**Coeff.**	**t-statistic**
Intercept	0.114	1.253	0.339	2.327**	0.397	2.314**	0.410	1.654*
POS_{MDA}	−0.033	−1.557	−0.042	−1.206	−0.028	−0.754	−0.084	−1.637
NEG_{MDA}	−0.007	−0.415	−0.004	−0.142	−0.012	−0.387	−0.013	−0.361
UNC_{MDA}	−0.022	−1.207	−0.032	−1.064	−0.048	−1.362	−0.101	−1.797*
LIT_{MDA}	−0.046	−1.370	−0.090	−2.159**	−0.104	−2.267**	−0.110	−1.695*
WMD_{MDA}	0.008	0.114	0.044	0.409	0.039	0.299	0.053	0.285
MMD_{MDA}	−0.037	−0.398	−0.050	−0.321	−0.063	−0.345	−0.109	−0.399
SMD_{MDA}	−0.037	−0.480	−0.013	−0.117	0.047	0.337	0.046	0.299
CNT_{MDA}	0.006	0.197	0.029	0.602	0.053	0.921	0.064	0.948
FOG_{RF}	0.001	0.364	0.000	0.090	0.000	−0.006	0.001	0.204
ΔNI_{it}	0.000	−0.025	0.010	0.337	0.017	0.540	0.018	0.415
$LOSSES_{it}$	−0.010	−0.548	−0.018	−0.767	0.001	0.046	−0.024	−0.651
$AVGACC_{it}$	−0.053	−0.315	0.146	0.522	0.181	0.573	0.579	1.204
LEV_{it}	0.029	1.069	0.063	1.479	0.077	1.732*	0.072	1.550
$BETA_{it}$	−0.018	−0.564	−0.064	−1.191	−0.063	−0.977	−0.050	−0.585
$STDSRV_{it}$	−0.498	−0.365	0.921	0.420	1.732	0.632	1.659	0.462
BTM_{it}	0.053	3.806***	0.072	2.730***	0.089	3.209***	0.129	3.380***
$SIZE_{it}$	0.002	0.313	0.002	0.280	0.000	−0.035	−0.001	−0.071
ICW_{it}	−0.011	−0.424	−0.051	−1.041	−0.049	−0.889	0.007	0.081
$RESTATE_{it}$	−0.021	−0.764	−0.038	−0.730	−0.048	−0.884	−0.081	−1.027
$VWRET_{it}$	0.167	1.229	0.104	0.496	−0.085	−0.333	0.058	0.173
$STDR_{it}$	0.099	0.070	−1.760	−0.591	−4.123	−1.345	−4.702	−1.145
$TVOL_{it}$	0.016	1.244	0.046	2.455**	0.063	2.836***	0.083	2.784***
Adjusted R^2		0.034		0.083		0.144		0.161

*, **, *** Denote 10 percent, 5 percent, and 1 percent two-tailed statistical significance levels, respectively.
n = 242. Year and industry fixed effects included. Standard errors clustered by firm.
Appendix A contains the definitions of all variables in this table.

V. CONCLUSIONS, IMPLICATIONS, AND LIMITATIONS

Cybersecurity risks continue to threaten enterprises, creating significant economic havoc (Lewis 2018). However, the literature and related market reaction research are still limited. This study fills the void by providing economically and statistically significant empirical evidence using various sentiment types. Our results from the main multivariate regression testing, robustness testing, alternate testing, sensitivity testing, and longevity testing offer affirmative answers to the three research questions that motivate this study. First, we apply sentiment analysis to RF and MDA disclosure separately. The statistical significance of the different sentiment types between RF and MDA and market reactions provides empirical evidence that the item-level disclosures are informative. Second, our model exhibits statistical significance and predictability after controlling for internal control weaknesses, financial restatements, and confounding events, suggesting that cybersecurity disclosures are informative. Third, and probably the most important result of this study, is the empirical evidence showing the significant associations of different sentiment types between RF and MDA with market reactions. The results are robust in the original sample set, the nonrepeating sample set, and the GM sample set in both short and long observation windows. In summary, this study not only answers the three research questions affirmatively but also contributes to the growing body of sentiment disclosure literature and future research in the following way.

Contributions

First, we are the earliest study to apply all eight sentiment types to risk disclosure textual analysis to examine their relationships with market reactions. The analysis (−1,1) shows that among the eight sentiment types in risk factor

TABLE 13

Regression Results of H2 in RF Sentiment Types with Genetically Matched Samples

Panel A: Short Windows

	CAR (−1,1)		*CAR* (−1,3)		*CAR* (−1,5)	
Variables	**Coeff.**	**t-statistic**	**Coeff.**	**t-statistic**	**Coeff.**	**t-statistic**
Intercept	0.215	3.106***	0.264	3.079***	0.237	2.555**
POS_{RF}	−0.002	−0.188	0.006	0.369	0.000	−0.016
NEG_{RF}	−0.007	−1.537	−0.004	−0.725	−0.004	−0.669
UNC_{RF}	−0.008	−1.246	−0.012	−1.368	−0.015	−1.485
LIT_{RF}	0.004	0.612	−0.001	−0.073	0.003	0.326
WMD_{RF}	−0.009	−0.971	−0.011	−0.950	−0.008	−0.597
MMD_{RF}	−0.016	−0.835	−0.027	−1.058	−0.018	−0.614
SMD_{RF}	−0.028	−1.189	−0.054	−1.685*	−0.069	−1.988**
CNT_{RF}	−0.002	−0.133	−0.010	−0.555	0.006	0.315
FOG_{RF}	−0.001	−0.754	−0.002	−0.971	−0.002	−0.844
$BREACH$	0.012	1.716*	0.016	1.728*	0.013	1.383
ΔNI_{it}	−0.015	−1.722*	−0.010	−0.908	−0.007	−0.556
$LOSSES_{it}$	0.007	0.998	0.005	0.545	0.010	1.016
$AVGACC_{it}$	0.018	0.327	0.044	0.620	0.057	0.776
LEV_{it}	0.007	0.591	0.013	0.894	0.014	0.903
$BETA_{it}$	−0.015	−1.247	−0.007	−0.446	−0.006	−0.353
$STDSRV_{it}$	−0.850	−1.364	−1.606	−1.957*	−1.503	−1.787*
BTM_{it}	0.014	1.635	0.032	2.822***	0.042	3.165***
$SIZE_{it}$	−0.003	−1.043	−0.003	−0.765	−0.001	−0.143
ICW_{it}	−0.013	−0.767	−0.016	−0.973	0.010	0.475
$RESTATE_{it}$	−0.020	−1.453	−0.030	−1.535	−0.028	−1.345
$VWRET_{it}$	−0.038	−0.579	−0.023	−0.262	−0.010	−0.111
$STDR_{it}$	−0.249	−0.741	0.219	0.404	0.047	0.069
$TVOL_{it}$	−0.005	−1.306	−0.009	−1.347	−0.009	−1.142
Adjusted R^2		0.155		0.117		0.127

*, **, *** Denote 10 percent, 5 percent, and 1 percent two-tailed statistical significance levels, respectively.
n = 266. Year and industry fixed effects included. Standard errors clustered by firm.
Appendix A contains the definitions of all variables in this table.

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disclosure, negative (*NEG*) is negatively and significantly associated with *CAR* (−0.018, p = 0.000). In contrast, weak modal words (*WMD*) are positively related and significant (0.028, p = 0.044). The model both shows meaningful explanatory power for *CAR* (Adjusted R^2 = 5.10 percent) and exhibits an economic significance of over $469 million in market capitalization based on *CAR*. In the MDA sentiment type analysis, we find that litigious (*LIT*) is negatively associated with *CAR*, with an economic significance of over $503 million. The model also exhibits a meaningful adjusted R^2 at 6.80 percent. Based on our samples, these results provide empirical evidence that other relevant and informative sentiment types exist beyond positive and negative.

Second, this study is the first to empirically test the differences in sentiment perception between risk factors and management's discussion and analysis disclosures. Our OLS regression testing of various sample groups validates that investors react to different sentiment types in other parts of the SEC reports, specifically between RF and MDA. The results provide more descriptive insights for management regarding the differences in investor sentiment perception between the different sections of SEC filing. In addition, these results have answered the first research question with empirical evidence that sectional disclosures, such as those in RF or MDA, are also informative to investors.

Third, this study extends the growing body of the literature (Amir et al. 2018; Frank et al. 2019; Jasa 2019; Li, No, and Wang 2018), supporting the informativeness of cybersecurity risk disclosure even in the longer observation windows. Our sensitivity analysis and the longevity tests show significant associations of *NEG* (−0.018, p = 0.000) and *WMD* (0.028, p = 0.044) with *CAR* in RF disclosure in the short window. At the same time, *LIT* and *UNC* demonstrate

TABLE 13 (continued)

Panel B: Long Windows

	CAR (−1,10)		*CAR* (−1,30)		*CAR* (−1,40)		*CAR* (−1,60)	
Variables	**Coeff.**	**t-statistic**	**Coeff.**	**t-statistic**	**Coeff.**	**t-statistic**	**Coeff.**	**t-statistic**
Intercept	0.113	0.979	0.093	0.582	0.043	0.232	0.135	0.522
POS_{RF}	0.007	0.392	0.048	1.691*	0.050	1.415	0.044	0.941
NEG_{RF}	0.004	0.532	0.007	0.610	−0.001	−0.050	−0.010	−0.549
UNC_{RF}	−0.030	−2.300**	−0.029	−1.495	−0.025	−1.070	−0.009	−0.266
LIT_{RF}	−0.004	−0.384	−0.015	−1.028	−0.021	−1.109	−0.003	−0.097
WMD_{RF}	0.009	0.530	−0.011	−0.411	−0.008	−0.238	0.000	0.008
MMD_{RF}	0.016	0.473	−0.014	−0.292	0.032	0.527	−0.080	−1.231
SMD_{RF}	−0.045	−1.085	−0.128	−2.204**	−0.150	−1.882*	−0.123	−1.105
CNT_{RF}	0.000	−0.013	−0.007	−0.262	−0.039	−1.029	−0.043	−0.762
FOG_{RF}	−0.002	−0.743	−0.001	−0.137	−0.003	−0.700	−0.002	−0.370
$BREACH$	0.022	2.079**	0.054	2.842***	0.067	2.779***	0.099	2.345**
ΔNI_{it}	0.007	0.404	−0.021	−1.062	−0.030	−1.117	−0.044	−1.495
$LOSSES_{it}$	0.002	0.155	−0.012	−0.650	0.002	0.086	−0.034	−0.997
$AVGACC_{it}$	0.103	1.213	0.411	2.272**	0.460	2.033**	0.695	1.788*
LEV_{it}	0.019	0.923	0.052	1.756*	0.063	1.886*	0.082	1.924*
$BETA_{it}$	−0.027	−1.369	−0.061	−2.050**	−0.050	−1.312	−0.012	−0.215
$STDSRV_{it}$	−1.210	−1.106	0.555	0.315	4.111	1.777*	2.631	0.777
BTM_{it}	0.045	3.042**	0.050	3.471***	0.049	2.320**	0.111	4.465***
$SIZE_{it}$	0.003	0.623	0.004	0.633	0.004	0.628	0.001	0.141
ICW_{it}	0.057	1.410	0.065	1.491	0.098	1.773	0.028	0.413
$RESTATE_{it}$	−0.023	−0.866	−0.009	−0.400	−0.010	−0.428	−0.042	−1.143
$VWRET_{it}$	0.028	0.224	−0.026	−0.134	−0.227	−0.862	−0.314	−0.948
$STDR_{it}$	0.405	0.353	0.678	0.372	−3.732	−1.461	−3.369	−1.230
$TVOL_{it}$	0.007	0.583	0.021	1.510	0.051	2.243	0.038	1.340
Adjusted R^2		0.010		0.236		0.252		0.283

*, **, *** Denote 10 percent, 5 percent, and 1 percent two-tailed statistical significance levels, respectively.
n = 266. Year and industry fixed effects included. Standard errors clustered by firm.
Appendix A contains the definitions of all variables in this table.

their associations with *CAR* even 60 days after the SEC report (−1,60) ($LIT = -0.110$, $p = 0.090$; $UNC = -0.101$, $p = 0.072$). As such, these results affirmatively answer the second and third research questions that disclosure is informative beyond just positive and negative; they also strongly predict market reactions in long windows.

Fourth, our results establish an empirical precedence associating sentiment types in RF disclosure to the various risk categories investors perceive. In addition to using keywords to identify different risk proxies (Campbell et al. 2014), future researchers may use sentiment types as risk proxies to analyze company performance and related market reactions.

Fifth, this study makes a solid methodological contribution by applying a rigorous statistical method that is less common in accounting research. This research is likely the first application of the PLS-SEM technique to textual and market archival data in accounting research. Furthermore, the causal predictive nature of the PLS-SEM approach (Hair et al. 2019b) and the PLSpredict (Shmueli et al. 2019) procedure validates and extends the significant associations noted under OLS to a causal prediction of market reactions by sentiment type. This study provides a foundation for researchers to expand the use of PLS-SEM in other accounting research.

Practical Implications

This study has practical implications for both management and regulation. For breached firms contemplating cybersecurity disclosure, management should be careful in applying sentiment types, particularly modal words, in their RF and MDA disclosures because of their considerable economic implications and enduring effects. The finding of the economic significance of sentiment types on cumulative returns and trading volumes provides a gauge of market

TABLE 14

Regression Results of H2 in MDA Sentiment Types with Genetically Matched Samples

Panel A: Short Windows

	CAR (−1,1)		*CAR* (−1,3)		*CAR* (−1,5)	
Variables	**Coeff.**	**t-statistic**	**Coeff.**	**t-statistic**	**Coeff.**	**t-statistic**
Intercept	0.092	1.912*	0.113	1.958*	0.112	1.826*
POS_{MDA}	0.002	0.167	−0.013	−0.826	−0.014	−0.873
NEG_{MDA}	−0.008	−1.351	−0.006	−0.805	−0.007	−0.872
UNC_{MDA}	−0.012	−1.710*	−0.013	−1.469	−0.017	−1.689*
LIT_{MDA}	−0.004	−0.495	0.009	0.830	0.009	0.756
WMD_{MDA}	0.019	0.829	0.023	0.836	0.028	0.880
MMD_{MDA}	−0.018	−0.594	−0.032	−0.836	−0.031	−0.725
SMD_{MDA}	0.044	1.407	0.066	1.543	0.037	0.772
CNT_{MDA}	−0.010	−0.749	−0.024	−1.419	−0.022	−1.196
FOG_{RF}	0.002	1.776*	0.002	1.086	0.001	0.558
$BREACH$	0.005	0.710	0.011	1.123	0.012	1.063
ΔNI_{it}	−0.014′	−1.605	−0.009	−0.884	−0.006	−0.533
$LOSSES_{it}$	0.009	1.359	0.006	0.661	0.010	0.983
$AVGACC_{it}$	0.007	0.132	0.044	0.677	0.056	0.797
LEV_{it}	−0.002	−0.163	0.005	0.350	0.008	0.490
$BETA_{it}$	−0.012	−0.854	0.001	0.081	0.004	0.201
$STDSRV_{it}$	−1.085	−1.765*	−1.850	−2.330*	−1.749	−2.080*
BTM_{it}	0.016	1.952*	0.034	3.049***	0.043	3.292***
$SIZE_{it}$	−0.002	−0.804	−0.002	−0.553	0.000	0.111
ICW_{it}	−0.017	−0.964	−0.028	−1.563	−0.003	−0.125
$RESTATE_{it}$	−0.018	−1.367	−0.028	−1.473	−0.025	−1.236
$VWRET_{it}$	−0.042	−0.662	−0.050	−0.583	−0.015	−0.158
$STDR_{it}$	−0.208	−0.610	0.175	0.325	0.089	0.126
$TVOL_{it}$	−0.004	−0.847	−0.007	−0.990	−0.007	−0.907
Adjusted R^2		0.142		0.114		0.111

*, **, *** Denote 10 percent, 5 percent, and 1 percent two-tailed statistical significance levels, respectively.
n = 266. Year and industry fixed effects included. Standard errors clustered by firm.
Appendix A contains the definitions of all variables in this table.

(continued on next page)

capitalization implications for management when drafting risk disclosures and management discussions. Appendix B provides examples of disclosures extracted from the samples tested to illustrate various sentiment types in the disclosures. Understanding how the investors will react to the different sentiments in the sections of the SEC reports increases companies' awareness of the possible implications of the words they use when drafting their disclosures. Furthermore, companies may run a sentiment analysis with the entire sentiment type to fine-tune their disclosure sentiment and possible market reactions before finalizing the disclosure draft.

The statistical and economic significance found in CAR and *TVOL* validates the disclosure's informativeness, as the SEC intended. However, the results should encourage the regulator to remain vigilant in requiring specific disclosure. Additionally, the SEC should strengthen the enforcement of timely disclosure by registrants regarding cybersecurity incidents. For example, only 17 of 423 (4 percent) cybersecurity incidents were reported in the current report (Form 8-K). In comparison, a majority (309 of 423 or 73 percent) of cybersecurity disclosures were first formally reported in registrants' annual reports (Form 10-K). The SEC can improve its timely filing objective through more stringent enforcement or severe penalties for failure to file timely reports after a cybersecurity incident (Richardson et al. 2019).

Research Limitations and Future Research Opportunities

This study has several limitations. First, it only includes U.S. public companies. Although some foreign companies listed on U.S. stock exchanges are among the breached firms, their filings are excluded from the final sample because of

TABLE 14 (continued)

Panel B: Long Windows

	CAR (−1,10)		*CAR* (−1,30)		*CAR* (−1,40)		*CAR* (−1,60)	
Variables	**Coeff.**	**t-statistic**	**Coeff.**	**t-statistic**	**Coeff.**	**t-statistic**	**Coeff.**	**t-statistic**
Intercept	0.016	0.222	−0.070	−0.670	−0.152	−1.213	−0.090	−0.525
POS_{MDA}	−0.018	−0.960	0.001	0.049	−0.019	−0.550	0.007	0.163
NEG_{MDA}	−0.010	−1.225	0.001	0.070	0.004	0.258	−0.018	−0.926
UNC_{MDA}	−0.009	−0.710	−0.007	−0.361	−0.014	−0.604	−0.023	−0.632
LIT_{MDA}	0.002	0.105	−0.010	−0.432	0.013	0.467	0.030	0.848
WMD_{MDA}	0.029	0.748	0.090	1.398	0.085	1.001	0.024	0.181
MMD_{MDA}	−0.028	−0.561	0.025	0.302	0.023	0.217	0.115	0.742
SMD_{MDA}	0.055	0.906	−0.069	−0.856	−0.020	−0.207	−0.022	−0.136
CNT_{MDA}	−0.024	−1.036	−0.012	−0.365	−0.038	−0.911	−0.001	−0.025
FOG_{RF}	0.003	1.249	0.003	0.910	0.002	0.501	0.003	0.560
$BREACH$	0.015	1.201	0.047	2.114**	0.061	2.260**	0.088	1.987**
ΔNI_{it}	0.006	0.340	−0.017	−0.877	−0.025	−0.868	−0.039	−1.255
$LOSSES_{it}$	0.002	0.195	−0.012	−0.602	0.001	0.042	−0.026	−0.742
$AVGACC_{it}$	0.099	1.131	0.392	2.267**	0.474	2.116**	0.708	1.772*
LEV_{it}	0.016	0.876	0.058	1.976**	0.065	1.887*	0.073	1.683*
$BETA_{it}$	−0.022	−1.023	−0.059	−1.966**	−0.044	−1.117	−0.007	−0.124
$STDSRV_{it}$	−1.292	−1.203	0.323	0.190	3.760	1.739*	2.553	0.768
BTM_{it}	0.046	3.148***	0.049	3.137***	0.050	2.288**	0.115	4.411***
$SIZE_{it}$	0.002	0.551	0.005	0.759	0.005	0.718	0.005	0.509
ICW_{it}	0.047	1.178	0.044	1.005	0.071	1.278	0.027	0.395
$RESTATE_{it}$	−0.020	−0.806	−0.013	−0.540	−0.013	−0.504	−0.041	−1.160
$VWRET_{it}$	0.018	0.147	0.024	0.128	−0.208	−0.816	−0.363	−1.148
$STDR_{it}$	0.293	0.266	0.963	0.540	−3.613	−1.367	−3.590	−1.215
$TVOL_{it}$	0.007	0.688	0.018	1.305	0.050	2.139**	0.042	1.424
Adjusted R^2		0.102		0.220		0.229		0.280

*, **, *** Denote 10 percent, 5 percent, and 1 percent two-tailed statistical significance levels, respectively.
n = 266. Year and industry fixed effects included. Standard errors clustered by firm.
Appendix A contains the definitions of all variables in this table.

a lack of consistent disclosures. Still, they are included in the GM samples. Since foreign companies are a significant part of the U.S. stock market,[18] the SEC should consider applying and enforcing the same cybersecurity risk reporting guidelines to U.S.-listed foreign companies. Future researchers can also investigate differences in sentiment based on industry.

Second, this study applies the word lists from Loughran and McDonald (2011) to assess the sentiment of risk factors and management's discussion and analysis disclosures. Although their sentiment word lists have not been explicitly developed for cybersecurity risks, they have been well-tested in accounting and finance disclosure literature, containing positive and negative corporate announcements and events. Therefore, there is no reason to believe that sentiment word lists do not apply to the cybersecurity risk context. However, researchers may consider testing new words specifically in the cybersecurity context to investigate the additional efficacy of sentiment capture in future disclosure studies.

Third, the current study does not break out the different breach types due to the inconsistency in the reported data. However, this would be an excellent area for future research if the breach-type data is more reliable and consistent. Similarly, our sample mainly consists of 10-Ks and 10-Qs due to the inconsistent nature of incident reporting using 8-Ks. With the new SEC (2018) guidance on cybersecurity disclosure, companies must report cybersecurity incidents in a more

[18] An untabulated calculation based on the market value of foreign owned companies found on the New York Stock Exchange (https://www.nyse.com/listings/international-listings) showed a combined value of $9.5 trillion or 31.7 percent of the total market value traded.

timely manner by filing 8-Ks. Therefore, the new SEC (2018) guidance would provide a good opportunity for researchers to review the disclosure content in 8-Ks in comparison to the empirical evidence of this study.

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APPENDIX A

Variable Definitions

Variables	Definition
Market Variables	
CAR_{it}	The three-day cumulative abnormal returns of firm *i* for fiscal year *t* during the three-day window [–1,1] surrounding the SEC report date closest to the last reported cybersecurity incident. Abnormal returns represent the difference between the raw return of firm $_i$ and the CRSP value-weighed market returns.
$TVOL_{it}$	Trading volume is measured by the natural logarithm of the average daily trading volume of firm *i* divided by the number of outstanding shares during the three-day window [–1,1] surrounding the SEC report date of the fiscal year *t* closest to the last reported cybersecurity incident, plus 1 to avoid a negative number.
$SPREAD_{it}$	The firm's average close bid-ask spread of firm *i* during the three-day window [–1,1] surrounding the SEC report date of the fiscal year *t* closest to the last reported cybersecurity incident, scaled by the averaged stock price over the same period × 100.
$AVGRET_{it}$	The filing returns is the natural logarithm of the average daily stock price of firm *i* during the three-day window [–1,1] surrounding the SEC report date of the fiscal year *t* closest to the last reported cybersecurity incident, scaled by the averaged stock price for the 60 trading day period ending two days before SEC report date.
$STDSRV_{it}$	The standard deviation of the daily return of firm *i* for the 60 trading day period ending two days before the SEC report date of the fiscal year *t* closest to the last reported cybersecurity incident.
$STDSRV_{it+1}$	The standard deviation of the daily return of firm *i* for the 60 trading day period starting two days after the SEC report date of the fiscal year *t* closest to the last reported cybersecurity incident.
Sentiment Variables	Type represents the disclosure section: RF = risk factors and MDA = management's discussion and analysis. All sentiment variables are multiplied by 100 for calculation.
$POS_{Type,it}$	The number of positive sentiment words based on LM word lists from SRAF divided by the total number of words in an item of a SEC report of firm *i* for fiscal year *t*.
$NEG_{Type,it}$	The number of negative sentiment words based on LM word lists from SRAF divided by the total number of words in an item of a SEC report of firm *i* for fiscal year *t*.
$UNC_{Type,it}$	The number of uncertain sentiment words based on LM word lists from SRAF divided by the total number of words in an item of a report of firm *i* for fiscal year *t*.
$LIT_{Type,it}$	The number of litigious sentiment words based on LM word lists from SRAF divided by the total number of words in an item of a report of firm *i* for fiscal year *t*.
$WMD_{Type,it}$	The number of weak modal sentiment words based on LM word lists from SRAF divided by the total number of words in an item of a report of firm *i* for fiscal year *t*.
$MMD_{Type,it}$	The number of moderate modal sentiment words based on LM word lists from SRAF divided by the total number of words in an item of a report of firm *i* for fiscal year *t*.
$SMD_{Type,it}$	The number of strong modal sentiment words based on LM word lists from SRAF divided by the total number of words in an item of a report of firm *i* for fiscal year *t*.

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APPENDIX A (continued)

Variables	Definition
$CNT_{Type,it}$	The number of constraining sentiment words based on LM word lists from SRAF divided by the total number of words in an item of a report of firm *i* for fiscal year *t*.
$FOG_{Type,it}$	The unreadability index is calculated as (words per sentence + percent of complex words) × 0.4 of firm *i* for fiscal year *t*.
Control Variables	
ANALYSTS	1 plus the natural logarithm of the number of analysts following firm *i* reported by I/B/E/S on the SEC report date of the fiscal year *t* closest to the last reported cybersecurity incident.
AVGACC	The absolute value of accruals is based on the difference between the net income (NI) minus the net cash flow from the operating activities (OANCF), deflated by total assets (AT) of firm *i* of the fiscal year *t* closest to the last reported cybersecurity incident.
ΔEST_{it}	The change in one-year-ahead mean EPS estimate reported in I/B/E/S deflated by the stock price of firm *i* at the end of the fiscal year *t* prior to the cybersecurity incident. The change represents the difference of the first estimate 30 days after the SEC report and the last estimate at least 30 days before the SEC report.
ΔNI_{it}	The change in net income before extraordinary items (NI) deflated by the lagged market value of equity of firm *i* at the end of the fiscal year *t* prior to the cybersecurity incident.
$BETA_{it}$	*BETA* is the firm *i*'s coefficient of the market based on the Fama-French three factors risk model with 250 days estimation window on the SEC report date of the fiscal year *t* closest to the last reported cybersecurity incident.
BREACH	A dichotomous variable for genetically matched samples: 1 for breached samples and 0 for matched unbreached samples.
BTM_{it}	The book-to-market ratio is defined as the book value of equity divided by the market value of equity of firm *i* measured at the end of the fiscal year *t* prior to the cybersecurity incident.
DNI_{it}	The net income before extraordinary items (NI) deflated by the lagged market value of equity of firm *i* at the end of the fiscal year *t* prior to the cybersecurity incident.
ICW_{it}	Internal control weaknesses as reported by Audit Analytics of firm *i* for fiscal year *t*: 1 for firms with one or more internal control weaknesss and 0 if no internal control weakness.
$INSTOWN_{it}$	Institutional ownership is the sum of shares of firm *i* owned by institutional investors as reported by Thomson Reuters scaled by total shares outstanding at the SEC report date of the fiscal year *t* closest to the last reported cybersecurity incident.
LEV	The debt leverage proxy is calculated using the sum of the long-term debt due in one year (DD1) and total long-term debt (DLTT) scaled by total assets (AT) of firm *i* at the end of the fiscal year *t* prior to the cybersecurity incident.
$LOSSES_{it}$	A dummy variable denoted with a value of 1 if firm *i* has one or more loss years over the previous five years and 0 if no loss reported over the previous five years.
$PREVOL_{it}$	The average natural logarithm of the average daily trading volume divided by the average outstanding shares of firm *i* for the 60 trading day period ending two days before the SEC report date of the fiscal year *t* closest to the last reported cybersecurity incident, scaled by the averaged stock price over the same period.
$RESTATE_{it}$	A financial restatement indicator reported by Audit Analytics: 1 indicates the firm *i* for fiscal year *t* has a restatement and 0 means no restatement.
$SIZE_{it}$	The natural logarithm of the market capitalization of firm *i* at the end of the fiscal year *t* prior to the cybersecurity incident. The market capitalization is calculated using the stock price at the end of the fiscal year multiple by the number of outstanding shares of common stock.
$STDR_{it}$	Standard deviation of daily returns of firm *i* over the designated observation windows *t* surrounding the SEC report date closest to the last reported cybersecurity incident.
VIX	An implied short-term market volatility index created by the Chicago Board of Options Exchange. VIX is calculated based on the put and call options of the S&P 500 within the targeted maturity of 22 trading days.
$VWRET_{it}$	The value-weighted stock returns of firm *i* for the 60 trading day period ending two days before the SEC report date of the fiscal year *t* closest to the last reported cybersecurity incident.

APPENDIX B

Examples of Disclosure Sentiment Types

Based on the cyber keywords found in the disclosure, we extract examples of such disclosures and highlight them with the different sentiment types in different formats as shown in the following.

AOL, 10K, RF, April 28, 2014 (second lowest ***POS***, third highest **[NEG],** fifth highest **UNC**)

On April 28, 2014, we announced the security **[incident]**. As a result of our **[investigation]** of the security **[incident]**, we have determined that there was **[unauthorized]** access to information regarding a significant number of user accounts. This information included AOL users' email addresses, postal addresses, address book contact information, encrypted passwords and encrypted answers to security **[questions]** that we ask when a user resets his or her password, as well as certain employee information. We **believe** that the individuals or entities have used this contact information to send "spoofed" emails that **appeared** to come from **roughly** 2 percent of our email accounts. We have no indication that the encryption on the passwords or on the answers to security **[questions]** was **[broken]**. In addition, at this point in the **[investigation]**, there is no indication that this **[incident]** resulted in disclosure of users' financial information, including debit and credit cards, which is also fully encrypted. We are working with ***best***-in-class external forensic experts and federal authorities to **[investigate]** this **[serious] [criminal]** activity.

The ongoing **[investigation]** of the security **[incident]** is at an early stage, and accordingly, no assurance can be given that the results of the **[investigation]** will not result in an **[adverse]** effect on our business, consumer **[dissatisfaction]**, **[damage]** to our brands, legal costs or liability, and a **[loss]** of consumers or revenues.

If our network and data security measures are **[breached]**, our products and services **may** be perceived as not being secure, consumers and customers **may [curtail]** or stop using our products and services, and we **may** incur significant legal and financial costs.

Aetna, Inc. 10K, MDA, February 17, 2017 (first in {**CNT**} and second highest in [LIT])

The HIPAA privacy [regulations] provide patients with rights to understand and control how their health information is used. States also have adopted [regulations] to implement provisions of the Financial Modernization Act of 1999 (also known as Gramm-Leach-Bliley Act ("GLBA")) which generally {**require**} insurers to provide customers with notice regarding how their non-public personal health and financial information is used and the opportunity to "opt out" of certain disclosures before the insurer shares such information with a non-affiliated third party. The GLBA [regulations] apply to health, life and disability insurance. Like HIPAA, GLBA sets a "floor" standard, allowing states to adopt more stringent {**requirements**} governing privacy protection.

The Cybersecurity Information Sharing Act of 2015 ("CISA") encourages organizations to share cyber threat indicators with the federal government and, among other things, directs HHS to develop a set of voluntary cybersecurity best practices for organizations in the health care industry. In addition, states have begun to enact more comprehensive privacy [laws] and [regulations] addressing consumer rights to data protection or transparency. States are also starting to issue [regulations] and proposed [regulations] specifically related to cybersecurity, such as the [regulations] issued by the New York Department of Financial Services. Complying with possibly conflicting cybersecurity **[**regulations**]**, which may differ from state to state, would {**require**} significant resources. In addition, differing approaches to state privacy and/or cyber-security [regulation] and varying enforcement philosophies may materially and adversely affect our ability to standardize our products and services across state lines. Widely reported large scale U.S. commercial data [breaches] increase the likelihood that additional data security [legislation] will be considered by additional states. These [legislative] and [regulatory] developments will impact the design and operation of our businesses, including the consumer business we are creating, our privacy and security strategy and our web-based and mobile assets.

APPENDIX C

Treated and Control Units Selection

In our dataset of 3,608 observations, we identified 133 companies that received data breach incidents (treated units). We use the genetic matching (GM) algorithm by Diamond and Sekhon (2013) to select a subsample of control units based on the standard industrial classification (SIC), book-to-market ratio (*BTM*), market capitalization (*MKTCAP*), and total assets (*TOTASSETS*) that are similar to the treated units. Our goal is to select 133 matching companies as a control group that shares similar characteristics and had the same probability of receiving a cyberattack but did not incur a data breach.

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APPENDIX C (continued)

Diamond and Sekhon (2013, 939) conducted a series of Monte Carlo experiments that showed the superior performance of the GM algorithm in terms of bias reduction and covariate balance. Compared to propensity score matching, the GM method's bias and root mean square error are 4.32 percent and 512, respectively, versus 51.2 percent and 1,832 of the latter. The main advantage is reaching a better balance between treated and control subsets compared to the Mahalanobis distance used in propensity score matching. The GM algorithm uses a generalized Mahalanobis distance formula that posits as:

$$d(\mathbf{X}_i\mathbf{X}_j) = \left\{(\mathbf{X}_i - \mathbf{X}_j)'\left(\sum^{-1/2}\right)'\mathbf{W}\left(\sum^{-1/2}\right)(\mathbf{X}_i - \mathbf{X}_j)\right\}^{1/2}$$

where $\mathbf{X} = [\text{SIC}, BTM, MKTCAP, TOTASSETS]$ is the matrix of the previously define firms' characteristics, $\sum^{-1/2}$ is the Cholesky factorization of the $\mathbf{X}$ covariance matrix, and $\mathbf{W}$ is a 4×4 positive semidefinite weight matrix. An artificial intelligent algorithm based on metaheuristics is used to compute the elements of W that minimize the distance between treated and control units.

Table C1 reports the statistics of the treated and control subsamples before and after matching. The genetic matching procedure achieved a fair balance between the two subsamples based on comparing the means of treated and control groups before and after matching.

TABLE C1

Genetic Matching Procedure: Balance Analysis

Variable	Before Matching			After Matching		
	Treated Mean [Median]		Control Mean [Median]	Treated Mean [Median]		Control Mean [Median]
SIC	5,940.5		4,733.0	5,940.5		5,826.3
	[6,020]		[4,813]	[6,020]		[6,020]
		(82.433)			(7.794)	
BTM	0.391		0.399	0.391		0.414
	[0.296]		[0.467]	[0.296]		[0.320]
		(−1.887)			(−5.032)	
MKTCAP	44,839		6,699	44,839		42,184
	[14,659]		[871.4]	[14,659]		[14,282]
		(46.97)			(3.269)	
TOTASSETS	111,981		16,019	111,981		117,175
	[12,961]		[1,100]	[12,961]		[12,001]
		(27.698)			(−1.499)	
Subsamples Size	133		3,475	133		133

Standard deviation mean difference in parentheses.
Appendix A contains definitions of all variables in this table.

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